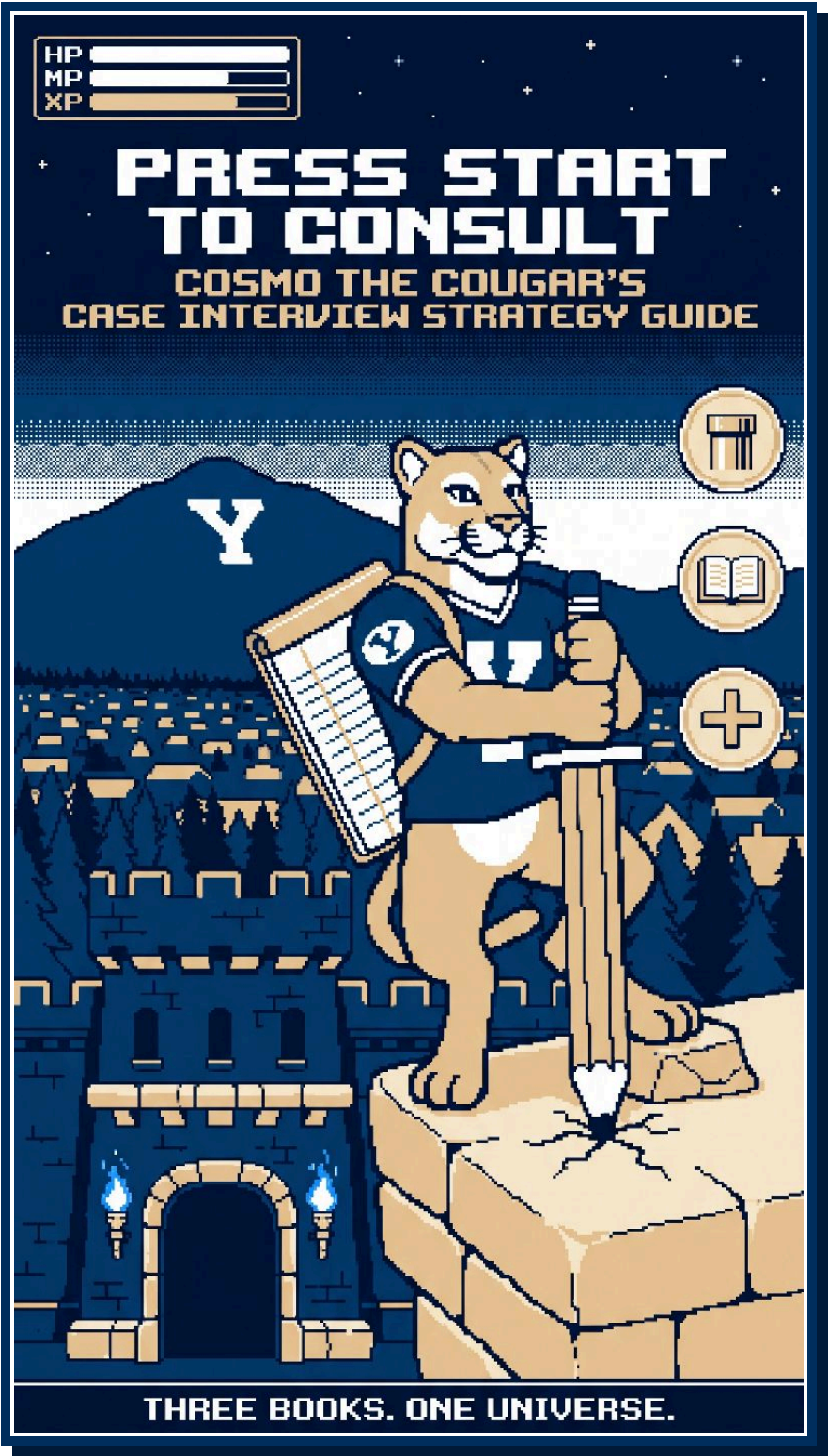




CONTENTS

SELECT A SECTION

THE WARP ZONE 3 PAGES

- 01 HUD Check
- 02 Spellbook
- 03 Overworld Atlas
- 04 Sector Codex
- 05 Quest Map in 60 Seconds
- 06 Boss Checklists

STORY MODE 10 PAGES

- 01 The Quest Line
- 02 Player Inventory
- 03 The Tutorial
- 04 Boss Encounters
- 05 Training Grounds

NEW GAME+ 18 CHAPTERS

- 01 Title Screen
- 02 The Quest Line
- 03 Player Inventory I: The Spellbook
- 04 Player Inventory II: The Overworld Atlas
- 05 Player Inventory III: Sector Codex, Healthcare
- 06 Player Inventory III: Sector Codex, Supply Chain and Operations
- 07 Player Inventory III: Sector Codex, Tech and Bonus Pages
- 08 Level 1: The Opening Cutscene
- 09 Level 2: The Quest Map
- 10 Framework Skill Trees
- 11 Level 3: The Sizing Dungeon
- 12 Boss Fight I: The Number Cruncher
- 13 Boss Fight II: The Idea Hydra
- 14 Final Boss: The Chairlift
- 15 Side Quests
- 16 Training Grounds
- 17 Trophy Room
- 18 Glossary

HOW TO USE THE SET

The three books share one universe. Cosmo is the player. The interviewer is Player 2, a co-op partner. The recruiting pipeline is the Quest Line. The case is the Boss Fight. A number without a "so what" is an empty treasure chest.

Cosmo the Cougar and the block Y are trademarks of Brigham Young University, used here for a student project. The running client, Wasatch Wheels, is fictional.

The Warp Zone

Press Start to Consult: the night-before cheat sheet.

BOOK 1 OF 3



01 HUD CHECK

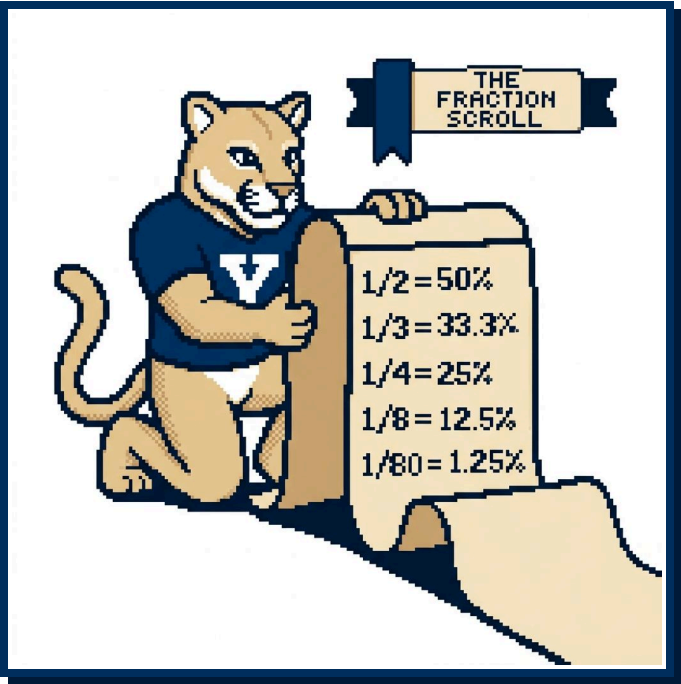
- 1 Opening Cutscene and Save Point (about 1 minute).** Take notes, then recap out loud: client, goal, numbers confirmed. Over video, 60 and 16 sound alike.
- 2 Dialogue Tree (2 to 3 minutes).** Ask three to five questions: business model, scope, target metric, gaps in your notes. Put a dodged question in your map.
- 3 Pause Menu, then Quest Map (2 minutes head down, about 3 to deliver).** Ask for two minutes each time. The coaches said any trained Player 2 expects it. Draw four regions with three quantified quests each, then renumber them by priority.
- 4 The variable middle.** Expect Number Cruncher rounds (about 2 minutes to structure, 3 to solve and share) and Idea Hydra rounds (3 to 4 minutes), in any number and any order.
- 5 Final Boss: The Chairlift (about 1 minute).** Ask for 30 seconds, then deliver a one-sentence recap, the recommendation before its support, the least-supported assumption, and next steps.

02 SPELLBOOK

- **Add:** line up the zeros, add from the left, say each subtotal out loud.
- **Subtract:** round the number you are taking away up to a clean figure, subtract, then add back what you over-subtracted: 500 minus 198 is 500 minus 200, plus 2, so 302.
- **Multiply:** strip the zeros, multiply the small numbers, put the zeros back: 400K units times \$150 is 4 times 15, then six zeros, \$60M.

- **Divide:** factor the divisor: 84K divided by 6 is half of 84K, then a third, about 14K.
- **Big numbers:** carry the unit (K, M, B, T). Thousands times thousands make millions.
- **Percents:** find 10% by moving the decimal, then scale: 15% is 10% plus half of it.

THE FRACTION SCROLL	
FRACTION	PERCENT
1/2	50%
1/3	33.3%
1/4	25%
1/5	20%
1/6	16.7%
1/7	14.3%
1/8	12.5%
1/9	11.1%
1/10	10%
1/11	9.1%
1/12	8.3%
1/15	6.7%
1/20	5%
1/25	4%
1/30	3.3%
1/40	2.5%
1/50	2%
1/80	1.25%
1/100	1%



The 1/80 trick. Give an 80-year lifespan an even spread and each year of age holds 1.25% of the population. People 65 and over span 15 years: 15 times 1.25% is about 19%. K-12 spans 13 grades: 13 times 1.25% is about 16%. The Atlas figures run lower (60M and 50M), so use the trick for a first pass and the Atlas for the final number.

THREE ROUNDING TRICKS.

☐

Round inputs to clean divisors and say so: Germany's 84M becomes 80M, and 3.5M Utahns at 3.0 per home become 1.2M households.

☐

Round one factor up and the other down so the errors offset: 48 times 21 becomes 50 times 20, or 1,000 (true: 1,008).☐

03 OVERWORLD ATLAS

Carry these in your head: US population, your city, the interview city, two countries or a continent, and your target firm's sector headline. Ask Player 2 for the rest.

WORLD AND US

FIGURE	VALUE
World population	8.2B
World GDP	\$120T
US population	340M
US households	130M
US average household size	2.5
US adults 18+	260M
US 65+	60M
US K-12 students	50M
US GDP	\$30T

UTAH	
FIGURE	VALUE
Utah population	3.5M
Utah households	1.2M
Utah average household size	3.0
Wasatch Front population	2.7M
Salt Lake County	1.2M
Utah County	720K
Provo-Orem metro	780K
BYU total enrollment	34K

REGIONS	
REGION	POP.
Asia	4.8B
Africa	1.5B
Europe	745M
European Union	450M
North America (US, Canada, Mexico)	510M

TEN BIGGEST US METROS	
METRO	POP.
New York metro	20M
Los Angeles metro	13M
Chicago metro	9.5M
Dallas-Fort Worth metro	8M
Houston metro	7.5M

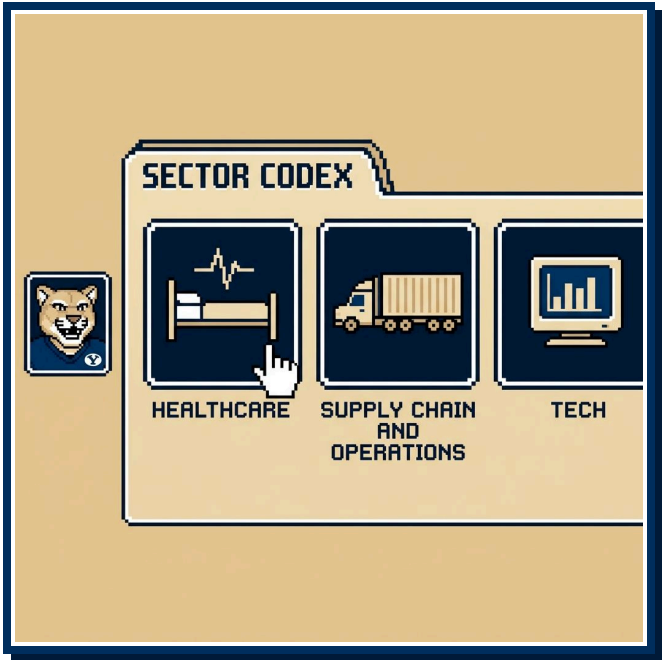
TEN BIGGEST US METROS, CONTINUED	
METRO	POP.
San Francisco Bay Area	7.5M
Washington DC metro	6.3M
Atlanta metro	6.3M
Miami metro	6.2M
Philadelphia metro	6.2M

TWELVE COUNTRIES	
COUNTRY	POP.
India	1.44B
China	1.41B
United States	340M
Indonesia	280M
Brazil	215M
Mexico	130M

TWELVE COUNTRIES, CONTINUED	
COUNTRY	POP.
Japan	124M
Germany	84M
United Kingdom	68M
France	68M
Canada	40M
Australia	28M

04 SECTOR CODEX

Load the page that matches your firm. A healthcare firm expects you to know Medicare enrollees cold.



HEALTHCARE	
METRIC	FIGURE
US healthcare spending	\$5.3T
Healthcare share of US GDP	18%
Spending per person	\$15.5K
US hospitals	6,100
Medicare enrollees	68M
Medicaid and CHIP enrollees	72M
Employer insurance covered lives	155M
Hospital operating margin	1% to 4%

SUPPLY CHAIN AND OPERATIONS	
METRIC	FIGURE
Logistics cost share of US GDP	9%
Trucking share of freight tonnage	72%
Truckload capacity (weight)	45,000 lb
Truckload capacity (pallets)	26 pallets
Truckload rate per mile	\$2.50
Ocean transit, Shanghai to LA	14 to 21 days
Last-mile share of total delivery cost	40% to 50%
Inventory carrying cost	20% to 30%
Fill rate target	95% to 98%
OEE world class	85%

TECH	
METRIC	FIGURE
Software gross margin	70% to 85%
LTV to CAC benchmark	3x
CAC payback benchmark	under 18 months
Net revenue retention (good)	100% to 120%
Annual gross churn, healthy SaaS	5% to 10%
Rule of 40	growth % + profit margin % >= 40
Marketplace take rate	10% to 30%
E-commerce share of US retail	16%

05 QUEST MAP IN 60 SECONDS

Build the four regions from this prompt, quantify each quest, and attach a "so that." Use your Compass, the hypothesis, to pick the region you start in.

- **Profitability:** Revenue (price, volume, mix), Cost (fixed, variable). Split each into external causes (demand, competitor moves) and internal causes (capacity, recent changes) with no overlap. Add what you suspect, such as a rival's price cut.
- **Market entry:** Market attractiveness (size, growth, share concentration), Customer (segments, willingness to pay, channel), Entry economics (price, margin, investment, payback), Capabilities and entry mode (build, partner, buy).
- **Growth:** Market, Product, Customer, Company. Or run the product-market matrix, current or new product by current or new market, and size each cell.
- **M&A:** Standalone value of the target, Synergies (revenue and cost, each sized), Deal economics (price against value, payback), Risks (culture, regulators, key people). Close with a binary answer: buy or pass.
- **Pricing:** Cost floor (unit economics), Customer ceiling (willingness to pay, elasticity), Competitor reference prices, Strategy and channel (skim or penetrate, dealer margin).
- **Operations:** Map the process, find the bottleneck, then Cost per unit, Capacity and utilization, Quality and waste, Speed (cycle time, on-time-in-full delivery).
- **The Sizing Dungeon, five floors:** (1) population, households, or businesses, (2) three or four segments, (3) percent who own or buy, (4) purchase frequency, under one per year for durables, (5) average price. True up 10% to 20% for fleets, rentals, and businesses. Market size means revenue per year unless Player 2 says otherwise. Utah e-bikes, all brands (the Wasatch Wheels case): 1.2M households, 72K on the Wasatch Front plus 12K rural for 84K, 14K a year, 16.8K with fleets, \$33.6M.

06 BOSS CHECKLISTS

BOSS FIGHT I: THE NUMBER CRUNCHER

- ☐ Recap the data, confirm each number, and name the unit: units or dollars.
- ☐ Write the equation in words before any number and get Player 2's nod.
- ☐ Solve out loud, one step at a time, and sense-check each result.
- ☐ Give the three-part answer: the number, the Loot (what it means and what the client should do), and what is next.

BOSS FIGHT II: THE IDEA HYDRA

- ☐ Recap the narrower question in 30 seconds, then ask for 30 to 60 seconds of prep.
- ☐ Name three necks that do not overlap (about 1 minute).
- ☐ Hang two to four heads on each neck.
- ☐ Say where you would start and what else (30 seconds).

FINAL BOSS: THE CHAIRLIFT

- ☐ Recap why the client called and what you found across the whole case.
- ☐ Recommend: a binary answer to a binary question, the support, and the least-supported assumption.
- ☐ Name three next steps, deeper or wider, starting with the one that firms up that assumption.

GAME OVER FIVE GAME OVER SCREENS

- ☐ You counted the installed base, a stock, instead of annual sales, a flow.
- ☐ You assumed 100% ownership, or any figure you cannot defend when Player 2 asks "why 60%?"
- ☐ You did the math in silence, or before the nod.
- ☐ You skipped the Pause Menu, or rambled past three minutes on the Quest Map.
- ☐ You handed over a number with no Loot, or called your dodged question dumb.

CONTINUE? ► YES NO

Each screen ends with **Continue?**, and the answer is one more live case out loud.



Story Mode

Press Start to Consult: the standard strategy guide.

BOOK 2 OF 3

01 THE QUEST LINE

HP  MP  XP 

You are Cosmo. The Boss Fight waits inside the Tanner Building, and the road to it runs through four stages: Network, Fit, Case, Offer. A former McKinsey engagement manager told the room that nobody grew up planning to be a consultant, so the field starts from zero.

Network is the Overworld. You travel, you meet NPCs at firm events and coffee chats, and the right conversation puts you in the interview chair. Then the firm walls the Overworld off: your resume and your handshake with a partner got you in, and they do not score the fight. Referral weight swings by firm. Some accept a partner's word, some prohibit it, and at some firms a note from a BYU MBA two years into the job carries weight.

Fit is Character Select: your backstory and your leadership stories from missions, projects, and jobs. Strong candidates have lost offers by grinding the case and skipping this screen, so rehearse your stories out loud the way you rehearse a sizing.

Case is the Boss Fight, the multi-phase encounter Player 2 scores.

Offer is the Victory Screen.

■ THE THREE WORLDS

World 0 is the gate. A recruiter screens you by phone. Speak in full sentences, laugh at the jokes, and expect a simple mini case if the caller comes from the business side. Many firms add a digital, gamified, or AI assessment here. As of 2026 it sits before the chair, and the firm uses your score to decide who gets one. Some firms now count that score inside the main process.

World 1 is first round: two 30-minute video cases with senior associates or managers two to four years past business school.

World 2 is the final: two or three 45-minute in-person cases with junior partners and partners four to twenty years in. A case alone runs about 30 minutes. With behavioral questions attached, plan on 45 to 60. One former McKinsey engagement manager ran ten minutes of chit-chat and 50 minutes of fit plus case.

■ FIVE BOSS FORMS

The mini case runs 15 to 20 minutes, often as a sizing question, in corporate roles and screening calls. The one-on-one full case runs 25 to 45 minutes. The digital, gamified, or AI assessment sits at World 0. In the presentation interview, an EY and PwC favorite, the firm hands you a blank six-to-eight-slide deck and

a ten-tab spreadsheet, leaves you alone in a room for three hours, then seats you before a panel. The group interview is rare for MBAs, and firms use it more for PhD candidates.

■ THE SCORECARD

Firms want three things from the fight: structure, problem solving, communication. Player 2 weighs three questions while you talk: whether your math and structure hold up, whether you take initiative the way an analyst does, and whether the team would enjoy working with you and managing you. Initiative looks like asking for the revenue data because you want to crack the problem, and saying why. Answer the third question well and a manager can send you alone to Omaha to sit with the cranky client.

Player 2 scores in a handful of big buckets, broad and qualitative. Weak math and weak structure show up anyway. Firms hire future partners, so Player 2 does not fail you for a slip on step 17 of a segmentation. One coach put it this way: above the bar beats the badge on your resume.

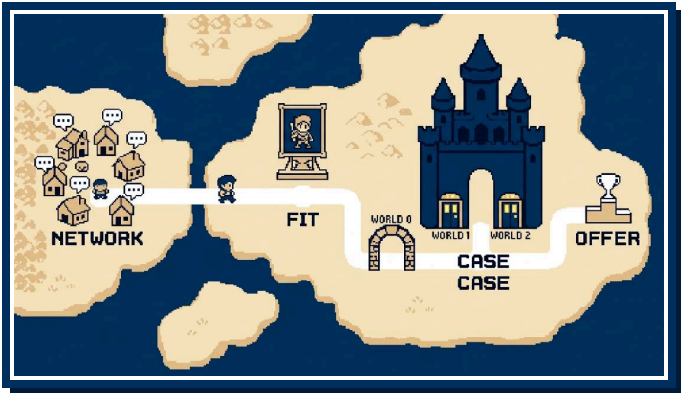
■ CO-OP MODE

Player 2 plays co-op. The coaches framed the interviewer as your partner in solving the case, and Player 2 scores collaboration. You narrate your thinking, and you check your equation with Player 2 before you compute, because work you do not say out loud earns no points. Narrate a wrong turn the same way: say the number looks high, say why, fix it, and move on, because Player 2 counts the catch in your favor.

■ BEYOND CONSULTING

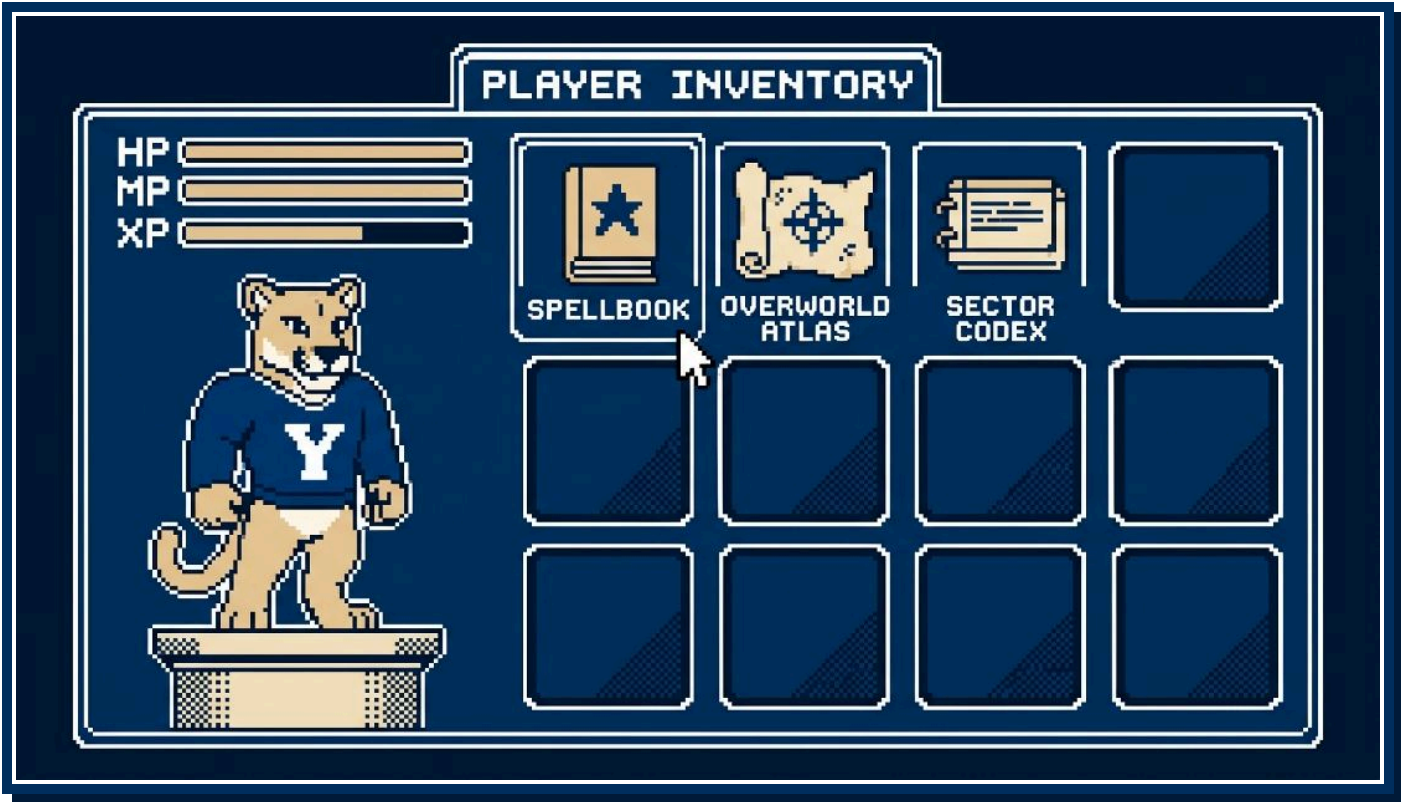
Casing has spread past the consulting firms into corporate strategy, general management, marketing, and finance roles. Capital One puts candidates through multiple mini and full cases. Bank of America runs mini cases. Delta and Walmart, recruiting on campus, told BYU that the MBA recruits needed stronger casing. Disney's corporate strategy group cases its candidates. Economic consulting firms load their cases with heavier math, because much of their work is litigation support: damages, antitrust, and securities cases.

Topics are random by design. A former McKinsey engagement manager put about a quarter of his firm's cases outside business, a police department that cannot recruit officers among them. The skill under test is breaking any situation into pieces you can solve.



02 PLAYER INVENTORY

Three items sit in the bag: the Spellbook for mental math (your MP), the Overworld Atlas for populations, and the Sector Codex for industry baselines.

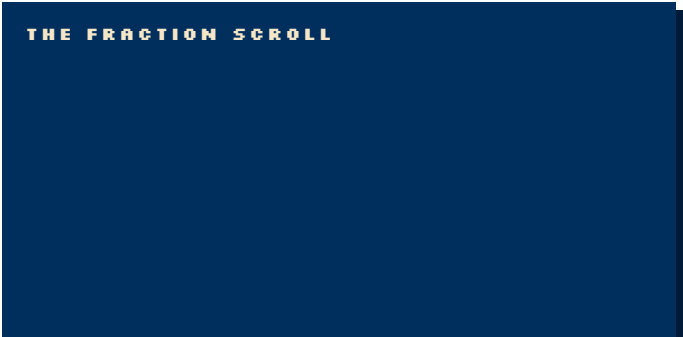


SPELLBOOK

Case math is sixth-grade arithmetic plus single-variable algebra and fractions. The six spells are add, subtract, multiply, divide, big numbers, and percents.

Cast by three rules. Write the equation before you touch a number and get Player 2's nod. Round to friendly numbers and say so. Solve one step at a time, narrating, and sense-check each result before you move on. On calculators, most firms say no, BCG puts your hands and paper on camera during video rounds, and Capital One allows them.

Memorize the Fraction Scroll so that you read 130 times 33% as one third of 130 and say about 43.



FRACTION	PERCENT
1/2	50%
1/3	33.3%
1/4	25%
1/5	20%
1/6	16.7%
1/7	14.3%
1/8	12.5%
1/9	11.1%
1/10	10%
1/11	9.1%
1/12	8.3%
1/15	6.7%
1/20	5%
1/25	4%
1/30	3.3%
1/40	2.5%
1/50	2%
1/80	1.25%
1/100	1%

The 1/80 trick. Assume an 80-year lifespan with the same number of people at each age. Each year of age is then 1/80 of the population, or 1.25%. People 65 and over span 15 years, so 15 times 1.25% is about 19%, and K-12 students span 13 grades, so 13 times 1.25% is about 16%. In the US that is 4.2M people per year of age. The trick runs about 10% above the atlas rows of 60M seniors and 50M students, so open with the trick and close with the atlas figure.



OVERWORLD ATLAS

The workshop set a need-to-know list: the US population, the city you live in, the city you are interviewing in, two countries or a continent, and, for a sector-focused firm, its headline figures. If a country is missing from your atlas, ask, and Player 2 will hand you the number.

WORLD AND US	
FIGURE	VALUE
World population	8.2B
World GDP	\$120T
US population	340M
US households	130M
US average household size	2.5
US adults 18+	260M
US 65+	60M
US K-12 students	50M
US GDP	\$30T

Household size sits at 2.5 for the country and runs larger in family-heavy regions like Utah. A former McKinsey engagement manager put Manhattan near 1.5. Carry the range as your rule: US household size runs between 1.5 and 4 depending on the place, so name the figure you picked and the reason you picked it.

Divide 340M by 2.5 and you land on 136M, above the atlas row. Take out the 8M Americans in group quarters such as dorms and care homes and you get 133M, and 2.5 is a rounded figure, so use the 130M row and move on.

UTAH BLOCK	
FIGURE	VALUE
Utah population	3.5M
Utah households	1.2M
Utah average household size	3.0
Wasatch Front population	2.7M
Salt Lake County	1.2M
Utah County	720K
Salt Lake City metro	1.3M
Salt Lake City (city proper)	210K
Provo-Orem metro	780K
Provo (city proper)	115K
BYU total enrollment	34K

TWELVE METROS: THE TEN BIGGEST, PLUS PHOENIX AND DENVER FOR THE MOUNTAIN WEST	
METRO	POPULATION
New York metro	20M
Los Angeles metro	13M
Chicago metro	9.5M
Dallas-Fort Worth metro	8M
Houston metro	7.5M
San Francisco Bay Area	7.5M
Washington DC metro	6.3M
Atlanta metro	6.3M
Miami metro	6.2M
Philadelphia metro	6.2M
Phoenix metro	5M
Denver metro	3M

TWELVE COUNTRIES	
COUNTRY	POPULATION
India	1.44B
China	1.41B
United States	340M
Indonesia	280M
Brazil	215M
Mexico	130M
Japan	124M
Germany	84M
United Kingdom	68M
France	68M
Canada	40M
Australia	28M

REGIONS

REGION	POPULATION
Asia	4.8B
Africa	1.5B
Europe	745M
European Union	450M
North America (US, Canada, Mexico)	510M

SECTOR CODEX

HEALTHCARE
US SPEND \$5.3T
MEDICARE 68M
HOSPITAL MARGIN 1% TO 4%

SUPPLY CHAIN AND OPERATIONS
9% OF US GDP
TRUCKING 72% OF TONNAGE
OEE 85% WORLD CLASS

TECH
SOFTWARE GROSS MARGIN 70% TO 85%
LTV TO CAC 3X
RULE OF 40: GROWTH + MARGIN >= 40

SECTOR CODEX

A healthcare firm expects you to carry Medicare enrollment the way a generalist carries the US population. Drop one of these into your structure and Player 2 stops explaining the sector to you.

HEALTHCARE

METRIC	VALUE
US healthcare spending	\$5.3T
Healthcare share of US GDP	18%
US healthcare spending per person	\$15.5K
Hospital care share of US health spending	30%
US hospitals	6,100
US hospital beds	920K
Typical hospital operating margin	1% to 4%
Average hospital length of stay	4.5 days
Medicare enrollees	68M
Medicaid and CHIP enrollees	72M
Employer-sponsored insurance covered lives	155M
Uninsured share of US population	8%

SUPPLY CHAIN AND OPERATIONS

METRIC	VALUE
US logistics cost share of GDP	9%
US logistics spend	\$2.5T
Trucking share of US freight tonnage	72%
Full truckload capacity (weight)	45,000 lb
Full truckload capacity (pallets)	26 pallets
Truckload all-in rate per mile	\$2.50
Ocean transit Shanghai to Los Angeles	14 to 21 days
Last-mile share of total delivery cost	40% to 50%
Inventory carrying cost per year (share of inventory value)	20% to 30%
Inventory turns: grocery	12 to 15
Inventory turns: general merchandise retail	4 to 8
Fill rate target	95% to 98%
OEE world class	85%

TECH

METRIC	VALUE
Software gross margin	70% to 85%
Hardware or marketplace gross margin	40% to 60%
LTV to CAC benchmark	3x
CAC payback benchmark	under 18 months
Net revenue retention (good)	100% to 120%
Annual gross churn: healthy SaaS	5% to 10%
Rule of 40	growth % + profit margin % >= 40
Magic number benchmark	0.75+
Burn multiple (good)	under 1.5
Marketplace take rate	10% to 30%
US e-commerce share of retail	16%

03 THE TUTORIAL



The first eight to ten minutes of the Boss Fight follow a script: Player 2 reads the prompt, you play it back, you ask a few questions, you ask for time, and you present a structure. That script holds across firms, so train it until it lives in muscle memory and your HP stays full for the middle, which changes with each case.

■ THE OPENING CUTSCENE

Player 2 reads two or three paragraphs, once. Write as you listen, in shorthand, and leave a wide right margin for the numbers you will confirm.

Then hit the Save Point. Play the prompt back out loud in your own words, in about a minute. Group the facts as you go, by segment and by number, so you have sorted the case before you touch a framework.

Over video, confirm each number one at a time: seventy and seventeen sound alike on a laggy call. Say the number, pause, and let Player 2 nod: that nod is your first save.

THE OPENING CUTSCENE

Two former ski-lift mechanics founded Wasatch Wheels in Ogden in 2012. The company employs about 900 people and sells three lines, commuter, recreation, and cargo, priced from \$1,500 to \$4,000.

■ THE DIALOGUE TREE

Three to five questions is the norm, in two to three minutes. You may ask none, but confirming a figure costs ten seconds and buys insurance. Player 2 dodges the novice branches: questions before the recap, "are there other objectives?", and data requests with no reason attached. Player 2 answers the advanced branches:

- Confirm the business model and scope. "Do the 1,200 dealers carry rival brands, or Wasatch alone?"
- Clarify the target metric. "Is there a revenue target or a timeline the client has in mind?"
- Ask for anything you missed. A number you did not catch is worth more than a clever question.

Phrase the question as a hypothesis and Player 2 hears a candidate with a Compass: "My hypothesis is that the 1,200 dealers carry rival brands alongside ours. Is that right?" Attach

the reason to any data request. "I would like margin by line so I can see which line should lead the plan" beats "Can I see margins?" and Player 2 hears hunger to solve the case.

Player 2 is there to clarify the business and the objective. A question that is analysis in disguise gets "we do not have that, but build it into your framework." A dodge means the question was good. Do not say "that must have been a dumb question." You will feel that line rising under pressure, and swallowing it is the easiest Game Over screen to skip.

Be brave about unfamiliar industries. If the case is about cargo insurance in Singapore, say so and ask how the money moves. One coach said he prefers to spend a minute explaining an industry over hiring an associate who nods along in the team room and learns nothing.

A short exchange after the prompt:

COSMO

Let me play that back. Wasatch Wheels, Ogden e-bikes, number two in North America by units. \$600M revenue, 400K units a year, 25 countries with 70% of revenue from the US. Three lines, sold through 1,200 dealers plus the website. Dana Okafor wants a growth strategy. Did I hear 25 countries and 70%?

PLAYER 2

You did.

COSMO

Two questions before I build. Does the client have a revenue target or a timeline in mind?

PLAYER 2

No target. She wants to see where the growth is and what you recommend.

COSMO

My hypothesis is that cargo earns the highest margin, since its price premium runs ahead of its build cost. Could you share margin by line, so I can weight my structure toward the line that earns the most?

PLAYER 2

We do not have that today. Build it into your structure and we may get there.

COSMO

Understood. I would like two minutes to organize my thoughts, then I will walk you through my approach.

PLAYER 2

Go ahead.

[COSMO WRITES FOR TWO MINUTES]

Player 2 dodged the second question, since margin by line is analysis you will do in Boss Fight I, and Cosmo moved on without an apology.

■ THE PAUSE MENU

Ask for the Pause each time, phrased as a plan the way Cosmo did above. Trained interviewers expect the request and say yes. One coach said he has watched hundreds of heads bent

over paper and remembers none of them, so the silence costs you nothing.

Use the two minutes on paper, in this order:

PAUSED

2:00

- ☐ One line at the top: the objective and the metric (revenue growth for Wasatch Wheels, and the client set no target).
- ☐ Four region names across the page, the first four that come, in shorthand.
- ☐ Three quests under each region, as fragments with a number in them: "TAM + growth by country, by line."
- ☐ Ten seconds to renumber the regions by priority and mark where you would start.

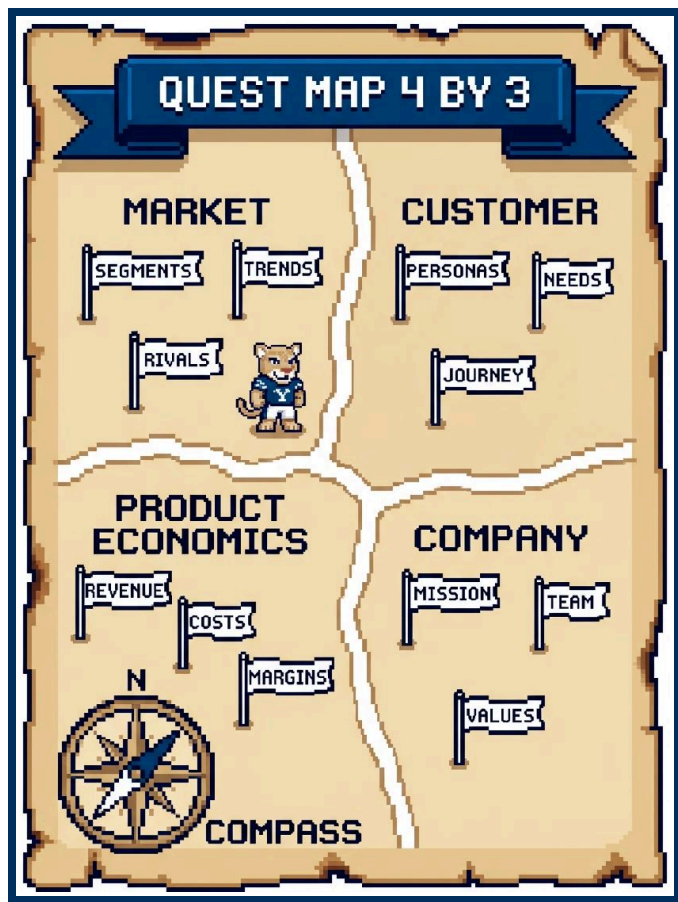
Successful McKinsey candidates in one coach's data averaged about 2:30 head-down and 2:15 to 3:00 presenting, and Bain runs snappier. Write fragments on the page, since you will speak the sentences.

■ THE QUEST MAP

A region is a workstream you would hand to a teammate on day one of a 4 to 12 week project. Miss a region now and the team discovers two weeks in that it needs a two-and-a-half-week consumer survey. In the interview you compress that plan into a 4 by 3 grid: four regions, three quests each.

Regions must be MECE: no overlapping regions, no unexplored map. If a quest could sit in two regions, pick one and move on. Framework building is an art, and Player 2 grades how easy your logic is to follow.

Each quest gets a number and a "so that." "Look at customers" is a wish. "Share of buyers in each rider segment and how it is shifting, so that we know which rider to build for" is a quest a teammate can run. The "so that" is the Loot you promise in advance. Player 2 marks a bare profit tree correct and forgets it, since a 12-year-old can recite "profit equals revenue minus cost." Keep the split and add what you suspect: "I suspect recreation is losing riders to cheaper rivals while cargo grows, so I want units and margin by line and dealer discounting on trail bikes."



Your Compass. A hypothesis at this stage is your best early guess at the answer, drawn from the prompt and the objective before you run a number. Wasatch Wheels wants growth, the US books 70% of revenue, and 25 countries split the rest, so start the guess there: growth sits abroad and in the commuter line. That guess sets region one. Spend the ten-second renumber moving the region that tests your Compass to the top of the page, and say why it goes there. State the Compass in one sentence after your last quest, and Player 2 gains a claim to steer you with and a reason to hand you data. Hold it as a guess. If the first analysis kills it, say so out loud, turn the Compass, and name the region you move to instead. Player 2 scores a hypothesis you stated and dropped with a reason above a case you ran with no compass at all.

The sample map for Wasatch Wheels growth, in priority order:

REGION	QUEST 1	QUEST 2	QUEST 3
1. Market	TAM and growth by country and by line, so that we chase the biggest pools	Share concentration in the top five markets, so that we know where share is winnable	Speed rules, subsidies, and tariffs by country, so that we skip closed doors
2. Customer	Penetration by rider segment (commuter, family hauler, trail) against potential, so that we find the biggest unmet demand	Willingness to pay versus the top two rivals, so that we know if price has headroom	Conversion by feature (range, battery, financing), so that we build what sells
3. Product economics	Contribution margin by line, so that growth adds profit as well as units	Margin by channel, dealer versus website, so that we know whether direct sales fund growth or cannibalize dealers	ROI by initiative (new country, new line, direct channel), so that we rank them
4. Company	Units per dealer across 1,200 dealers, so that we know if the channel can carry more	Assembly capacity against 400K units, so that we can ship what we sell	Free cash flow and payback, so that we fund the plan we pick

Your Compass: growth comes from the commuter line in countries where e-bikes are common. Start in Market, quest 1.

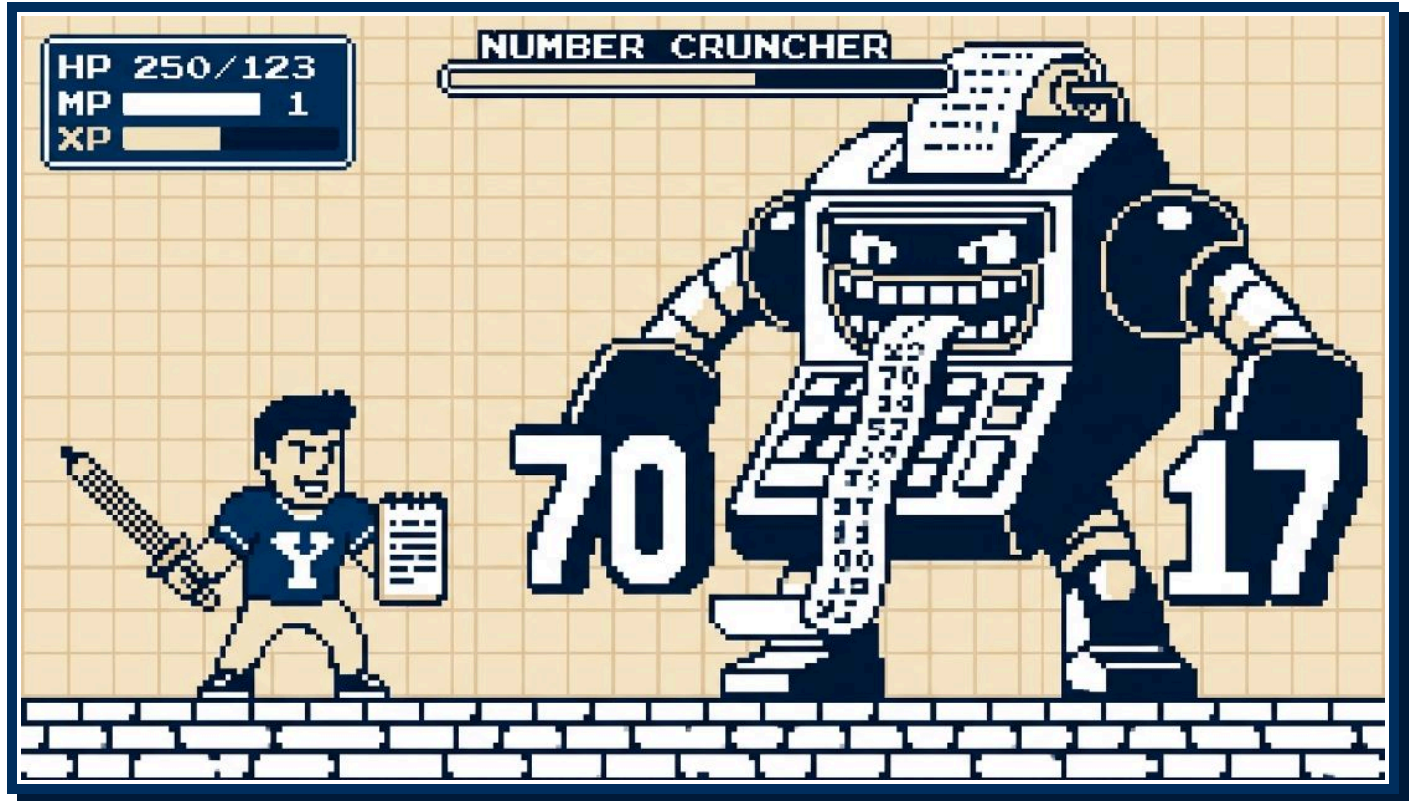
Deliver the map in about three minutes. Say the headline first: "I see four areas, and I would start with the market." Name the regions in priority order, give one sentence per quest with the

number and the "so that," and close with where you would start and the reason. If you run past three minutes, you are repeating a quest or reading the page aloud without priorities. One coach admitted a four-minute candidate once got the job, then told the room to train for three.

04 BOSS ENCOUNTERS

Between the Quest Map and the close, Player 2 throws quant and creative questions at you in any order. You beat each with the same habit: structure first, talk through the work, end with the Loot.

■ BOSS FIGHT I: THE NUMBER CRUNCHER



The Number Cruncher has three forms. In verbal form, Player 2 reads you a paragraph of numbers. In chart form, you get a table such as a P&L. In graph form, you get a bar or line chart, and you read the title, axes, and units before a single bar. Your process stays the same.

- 1 Recap.** Say what you are solving for, then read the data back by segment. Confirm each figure.
- 2 Structure.** Lay out the equation in words before you touch a digit. Ask for missing inputs and say why you want them. Then stop and get the nod from Player 2, so you catch a missing step before you burn three minutes on the wrong equation.
- 3 Solve.** Work each line out loud, round to friendly numbers, and sense-check as you go.
- 4 Loot.** Give the three-part answer: the number, what it means for the client, and what comes next.

Plan on two minutes to structure and three to share. Player 2 often stops you after the setup, since full arithmetic takes three to four times as long. Math is the one part of the case with a

single right answer, so a slip shows. Catch it and fix it out loud, without an apology spiral.

Reading an exhibit. Player 2 slides a table across the desk.

Wasatch Wheels revenue by product line, in \$M

PRODUCT LINE	LAST YEAR	THIS YEAR
Commuter	300	330
Recreation	200	180
Cargo	100	150

Read it in this order before you speak: title, axes, units, period, the biggest bar, the trend, the outlier, the so what. Then work it out loud. Totals first: \$600M to \$660M, +10% overall. Growth by line next: Commuter +10%, Recreation -10%, Cargo +50%. Mix last: Cargo mix rose from 16.7% to 22.7%, and you read 16.7% off the Fraction Scroll, since 100 of 600 is one sixth. Recreation is the outlier, since it is the one line that shrank.

LOOT

The Loot: growth comes from cargo while recreation shrinks, and margin by line is the next question before money follows cargo.

The Sizing Dungeon. A market sizing question has five floors: the base (population, households, or businesses), three or four segments, the share who own or buy, purchase frequency (below one per year for a durable good), and average price. Market size means revenue dollars per year unless Player 2 says otherwise. Then add a 10% to 20% true-up for buyers your model skipped.

Player 2 asks how many e-bikes sell in Utah each year, in units and dollars.

FLOOR	ASSUMPTION	CALCULATION
Base	Utah population 3.5M, household size 3.0	1.2M households
Segments	Wasatch Front 75%, rural Utah 25%	900K and 300K households
Ownership	8% of Wasatch Front homes, 4% of rural homes	72K plus 12K = 84K e-bikes
Frequency	One purchase per 6 years	84K divided by 6 = 14K units per year
True-up	Fleets and rentals add 20%	14K times 1.2 = 16.8K units per year
Price	Average \$2,000	16.8K times \$2,000 = \$33.6M per year

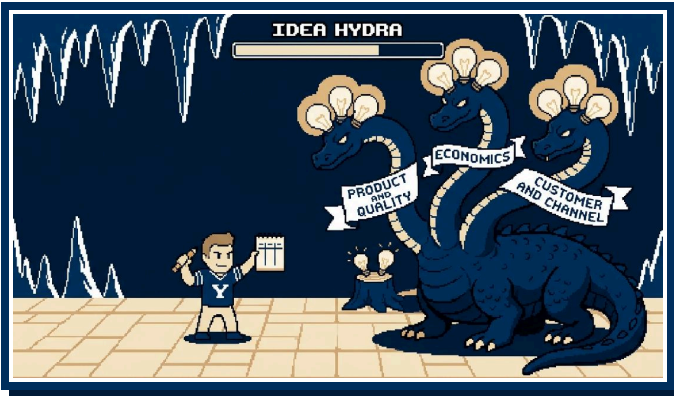
GAME OVER

Game Over: you announce 84K as the answer. That figure is the installed base, a stock, and Player 2 asked for sales per year, a flow. **Continue?** Divide by the replacement cycle.

Sense-check before you announce: Utah holds about 1% of the US population, so scale 16.8K units to a US market near 1.7M a year. The client sells 400K units and books 70% of revenue at home, so about 280K US units, a sixth of that market and a believable share. Then give the three-part answer. The number: about 16.8K units and \$33.6M a year. The Loot: Wasatch Wheels books \$600M in revenue, so the whole Utah market is under 6% of the company even at 100% share, and growth has to run through bigger markets. Next: pull dealer sell-through, the units dealers sold to riders, for the client's Utah share, then size California and Texas.

If Player 2 asks why 8%, say Wasatch Front households skew toward commuters and trail riders who can afford a \$2,000 bike. **TROPHY UNLOCKED: REASONABLE NUMBERS.**

BOSS FIGHT II: THE IDEA HYDRA



The Idea Hydra is the creative question. Cut one head and two more appear, so you win by sorting the heads onto three necks. The fight is a miniature Quest Map on a narrower question and takes three to four minutes.

BOSS FIGHT II: THE IDEA HYDRA

- ☐ Recap and confirm the question, 30 seconds.
- ☐ Ask for 30 to 60 seconds of prep. Player 2 expects the ask. Use the silence to pick three categories.
- ☐ Fill each category with two to four points.
- ☐ Close with where you would start and what else you would add, 30 seconds.

The Wasatch Wheels version: a national big-box retailer has launched a private-label e-bike at 40% below the client's average price, and dealers want price support. Player 2 asks what you would investigate about the retailer's move before deciding how to respond.

Neck one, product and quality: specs, battery, warranty, and who builds it, since a knockoff with no service network earns one sale per customer. Neck two, economics: whether the retailer sells at a loss to pull store traffic, how long it can hold that price, and who supplies it. Neck three, customer and channel: which segment is switching, how much brand matters here, and what your 1,200 dealers see on the floor.

Start with a mystery shop of the product and a pull of dealer sell-through data for the last 90 days. Then name two responses to test before you touch price: a dealer-exclusive model and a financing offer.

PLAYER 2

Dana Okafor is riding the chairlift with me this afternoon. One ride, about a minute. She wants to know what we recommend on growth. Tell me what I should say to her.



COSMO

Could I take 30 seconds to organize my notes before I answer?



[COSMO WRITES FOR 30 SECONDS]

COSMO

Tell her this. She brought us in to build a growth plan for a \$600M brand. We sized her home market, read the product lines, and worked the private-label threat. I opened expecting the commuter line abroad to lead, and I turned once I read the line results. Three things. First, the recommendation: grow outside Utah and behind cargo. Utah takes about 16.8K e-bikes a year across all brands, \$33.6M, so the home state cannot carry the plan, and cargo grew 50% this year while recreation fell 10%. Hold off on a price war with the private label until dealer sell-through is in. Second, the number I trust least is the 8% of Wasatch Front households I assumed own an e-bike. Take it to 4% and Utah drops to about 9.6K units and \$19.2M, so I would firm it up with a consumer survey. Third, next steps: margin by line before we fund cargo, the dealer sell-through pull, and California and Texas sized the way we sized Utah.



FINAL BOSS: THE CHAIRLIFT



The wrap-up prompt is corny and unmistakable: Player 2 has dinner with the CEO tonight, or in this case shares her chairlift. Ask for 30 seconds to organize your notes. A trained interviewer grants it.

You take three steps on the ride: recap the problem and what you did, recommend, and give next steps, meaning three areas to push deeper or wider. Answer a binary question, such as whether to match the private-label price, with the binary answer first, then next steps. Answer an open question, such as how to grow, with the recommendation first. Either way, name the number you trust least and say how you would firm it up, such as a consumer survey on price sensitivity before you raise prices. Draw the recommendation from the whole case.

Player 2 will push on one point. Hold or move, with a reason either way.

TROPHY UNLOCKED: ANSWER FIRST.

05 TRAINING GROUNDS



Casing is a decathlon: you train the weak events and let the strong ones coast. Rate yourself by section after each live case and spend the week on the lowest score.

The only XP that counts comes from out-loud reps with a partner. Math in silence is easy. Math with a person watching and taking notes is a different skill, and it is the one on the scoresheet. An MBA second-year with 20 to 30 cases behind them, given and taken, will tell you where you come off flat. AI casing works for drills, structure, and concept review. Its feedback runs positive, because the model praises whatever you did, and that is dangerous in the week before World 1.

FOUNDATIONS

Watch and work through five full live cases. Learn the business basics you skipped (a P&L, fixed versus variable cost) and skim industry primers so a hospital or trucking case does not start from zero.

INTENSIFY

Master the core frameworks and start blending them, since few prompts fit one tree. Complete ten out-loud cases with a partner or coach. Practice your Character Select story until you can tell it in two minutes without notes.

ADAPTABILITY

Get your mental math fast and accurate, get comfortable reading charts under a clock, and sharpen the hydra. Take cases in different styles. A Bain case runs snappier than a McKinsey case, and a Capital One mini case is its own creature.

The BYU MBA Strategy and Consulting Association runs case practice on Tuesdays at 5 pm. Show up, take a case, give a case. Trophies like the two above mark the habits worth repeating, one unlock per habit, and New Game+ carries the full set of fifteen. You earn the XP for the Dungeon in that room.



New Game+

Press Start to Consult: the 100% completion tome.

BOOK 3 OF 3

01 TITLE SCREEN

You are Cosmo, and a chair in the Dungeon has your name on it.

■ THREE BOOKS, ONE UNIVERSE

The Warp Zone is the three-page cheat sheet for the night before an interview. Story Mode is ten pages, enough to run a full case with a partner. New Game+ is the completion run. Read it front to back before your first live case, then treat it as a reference.

One coach framed the whole exercise as a selection test that runs both ways: the case compresses the job into thirty minutes, so if you enjoy this kind of problem solving you will enjoy the work. Dread the half hour, and you have learned something about the next two years before you sign anything.

■ HP, MP, XP

HP is composure. It drains when a number comes out wrong on camera or Player 2 pushes back. You refill it by asking for the Pause Menu and rounding to a friendly number out loud.

MP is mental math capacity. Spells cost MP, and a tired caster fumbles a 12.5% that a rested one would land. You grow the bar with the ten-minute daily drill in the Spellbook.

XP is live cases completed out loud with a partner. Reading this tome earns zero XP. One coach put it this way: a flawless case delivered to your own shoes earns no offer.

■ CO-OP, NEVER PVP

The workshop coaches framed Player 2, the interviewer, as your partner in the case. Firms hire the person they want in the team room at 11 pm, so play like that person: think out loud, say why you want each piece of data, and treat pushback as a teammate testing the plan.



02 THE QUEST LINE

The Quest Line has four stages: Network, Fit, Case, Offer. The Overworld gets you into the chair, Character Select tests your backstory, the Boss Fight tests your structure, math, and judgment, and the Victory Screen follows.

FIT VERSUS CASE

Character Select is the fit interview, or the PEI at McKinsey. A former McKinsey engagement manager told the room that strong candidates have lost offers by over-indexing on the case and under-indexing on fit. The same coach added that nobody grew up planning to be a consultant, so you learn both halves from zero. Split your prep time between them.

THE THREE WORLDS



**WORLD 0:
THE GATE**



**WORLD 1:
TWO VIDEO CASES,
30 MINUTES EACH**



**WORLD 2:
TWO OR THREE
IN-PERSON CASES,
45 MINUTES EACH**





**VICTORY
SCREEN**

WORLD	FORMAT	PLAYER 2	TIMING AND NOTES
World 0 (the gate)	Recruiter screen, then a digital assessment	A recruiter, then software	Short call, sometimes with a mini case
World 1	Two video cases	Senior associates or managers, 2 to 4 years out of business school	30 minutes each
World 2	Two or three in-person cases	Junior partners and partners, 4 to 20 years in	45 minutes each

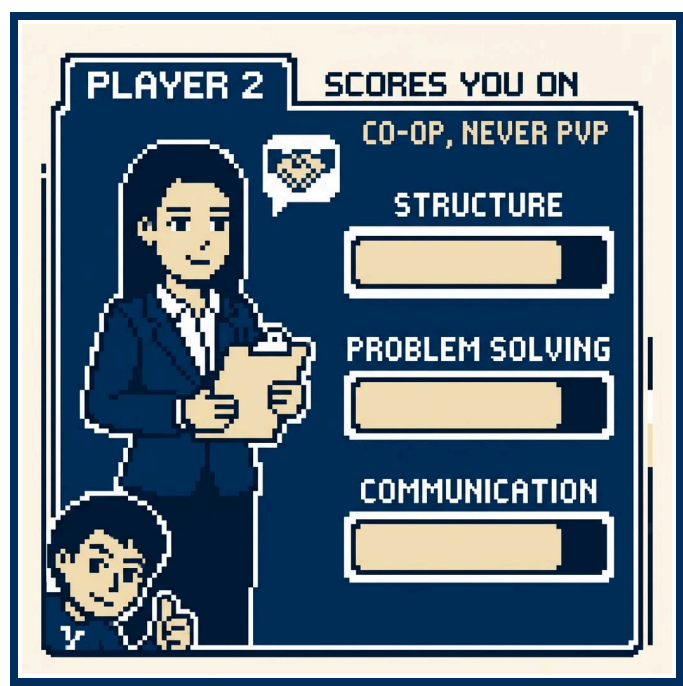
A case runs about 30 minutes, and 45 to 60 with behavioral questions attached. One coach described his McKinsey format: 10 minutes of chit-chat, then 50 minutes for behavioral questions and the case.

FIVE BOSS FORMS

- 1 **Mini case.** It runs 15 to 20 minutes and shows up in corporate roles and screening calls. Market sizing is its usual prompt.

- 2 **One-on-one full case.** It runs 25 to 45 minutes, and a coach dated the format to the 1930s. You get a prompt, clarifying questions, the Quest Map, some quant, some creative, and a close.
- 3 **Digital, gamified, or AI assessment.** It sits before the chair as of 2026, though some firms have moved it inside the main process, and past that bar it counts for little. The full-case skills carry over.
- 4 **Presentation interview.** One coach described the EY and PwC version: a blank deck of 6 to 8 slides, a spreadsheet with ten tabs, three hours alone in a room, then a presentation to a panel. The hours vary by firm and the shape stays.
- 5 **Group interview.** Firms use it for PhDs more than MBAs: about seven people share a small room and one case for two hours while the panel watches them work as a team. Expect to skip it.

■ THE RUBRIC BEHIND PLAYER 2



Firms want three things: structure, problem solving, communication. Behind the rubric sit three questions. The first asks whether you can reason in structures and numbers, and Player 2 reads that off your framework and your math. The second asks about initiative. You show it when you ask for the revenue data because you want to solve the problem, the way an analyst would. The third asks whether people would enjoy working with you and managing you: you are the associate a partner can send to Omaha to sit with the cranky client.

Scoring is broad and qualitative, a handful of big buckets, and weak math or structure shows through them. Firms are hiring future partners, so a slip on step 17 of a segmentation does not decide the outcome. Above the bar beats the badge on your resume.

■ CASING BEYOND CONSULTING

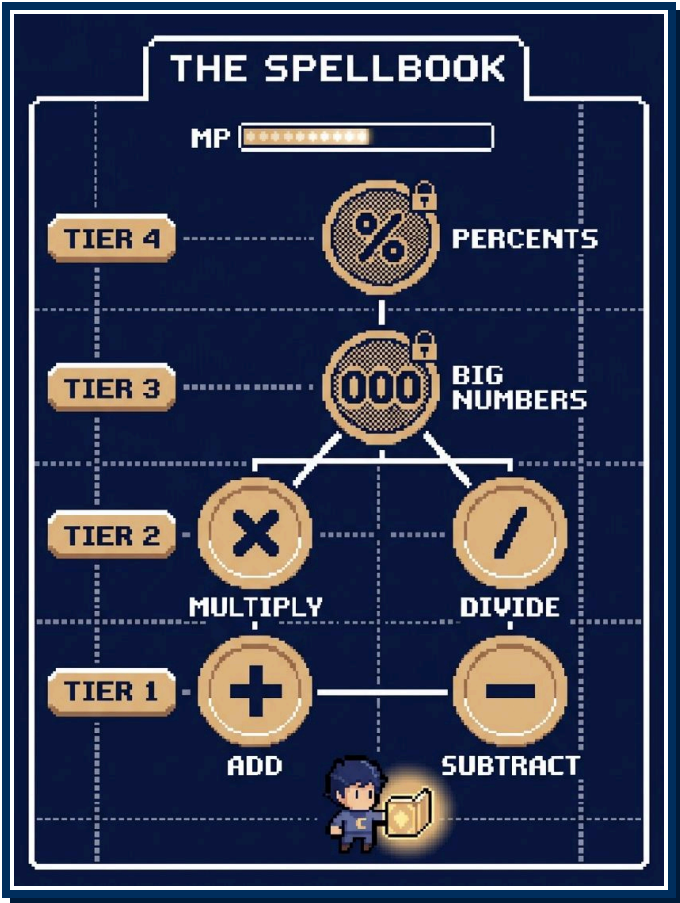
Corporate strategy, general management, marketing, and finance roles case candidates too, and more of them do it each year. Capital One requires multiple mini and full cases, and Bank of America uses mini cases. Delta and Walmart told BYU they wanted stronger casing from MBA recruits, and Disney's corporate strategy group cases its hires. Economic consulting firms run heavier math, because their work is forensic, such as a damages figure for residents after 15 years of pollution. The field runs past the named firms into hundreds of boutiques, and a firm's niche sets how quantitative its cases run. Read the niche before you prep: a shop built on litigation or pricing work asks for more arithmetic than a generalist strategy shop.

■ THE RANDOMNESS

The former McKinsey engagement manager listed his own interview cases: floating harbors for disaster relief, retail pricing promotions in Canada, a philanthropy chasing younger donors, cost overruns at an automaker, and CPG soap. He put about a quarter of his firm's cases outside business, such as a police department that cannot recruit officers. The randomness is the point: Player 2 hands you a situation you have not seen and watches you break it into pieces and build it back up.

03 PLAYER INVENTORY I: THE SPELLBOOK

Your MP bar measures the arithmetic you can do out loud while Player 2 watches, and the six spells stop at sixth-grade math, single-variable algebra, and fractions. Silent math at your desk is easy, and math with a witness burns through MP, so run the drills below to grow the bar.



THE SKILL TREE

Unlock the spells in order, since each leans on the ones below it, and each carries a technique and a drill.

Add (Tier 1). Work left to right from the biggest place value. Wasatch Wheels' product lines this year sum to 330 plus 180 plus 150: the hundreds give 500, the rest give 160, total \$660M. Drill: add three numbers from any exhibit out loud without writing the carry.

Subtract (Tier 1). Count up from the smaller number instead of borrowing. Recreation fell from 200 to 180, a gap of 20, and 20 over 200 is 10%, so the line moved -10%. Drill: read two columns of any table and call the change on each row.

Multiply (Tier 2). Split one factor into pieces you can hold. 16.8 times 2 is 32 plus 1.6, so 16.8K units times \$2,000 is \$33.6M. Drill: two-digit times one-digit for a week, then two-digit times two-digit.

Divide (Tier 2). Swap the division for a fraction you know. 84K e-bikes over 6 years is 14K per year because 6 times 14 is 84. Drill: divide random three-digit numbers by 3, 6, 7, 8, and 12 out loud.

Big Numbers (Tier 3). Strip the units, compute with the small numbers, then restore the units with the K/M/B/T rules below. Drill: five multiplications a day that cross a unit boundary.

Percents (Tier 4). Anchor on 10%, then scale. Drill: take percents of Overworld Atlas numbers, 12% of 340M, 15% of 130M, until you answer on reflex.

THE FRACTION SCROLL

THE FRACTION SCROLL

$\frac{1}{2} = 50\%$	$\frac{1}{12} = 8.3\%$
$\frac{1}{3} = 33.3\%$	$\frac{1}{15} = 6.7\%$
$\frac{1}{4} = 25\%$	$\frac{1}{20} = 5\%$
$\frac{1}{5} = 20\%$	$\frac{1}{25} = 4\%$
$\frac{1}{6} = 16.7\%$	$\frac{1}{30} = 3.3\%$
$\frac{1}{7} = 14.3\%$	$\frac{1}{40} = 2.5\%$
$\frac{1}{8} = 12.5\%$	$\frac{1}{50} = 2\%$
$\frac{1}{9} = 11.1\%$	$\frac{1}{80} = 1.25\%$
$\frac{1}{10} = 10\%$	$\frac{1}{100} = 1\%$
$\frac{1}{11} = 9.1\%$	

THE FRACTION SCROLL

FRACTION	PERCENT
1/2	50%
1/3	33.3%
1/4	25%
1/5	20%
1/6	16.7%
1/7	14.3%
1/8	12.5%
1/9	11.1%
1/10	10%
1/11	9.1%
1/12	8.3%
1/15	6.7%
1/20	5%
1/25	4%
1/30	3.3%
1/40	2.5%
1/50	2%
1/80	1.25%
1/100	1%

Case math looks like 130 times 33%: read 33% off the scroll as a third, and a third of 130 is about 43. Run it in reverse for division: 84K over 6 is 84K times 16.7%, so 14K. Memorize the scroll until you say 14.3% on sight of 1/7. One coach put it this way: with the scroll memorized, you look sharper than you are.

■ THE 1/80 TRICK

The atlas lists US life expectancy at 78. Round it to 80 and assume each year of age holds the same share of people. One year of age is then 1/80 of the population, 1.25%, or 4.2M people:

- People 65 and over: 80 minus 65 leaves 15 years, and 15 times 1.25% is about 19%.
- K-12 students: 13 grades times 1.25% is about 16%.

The atlas figures are 60M and 50M, so both estimates land within about 10%, closer than Player 2 needs from a two-second trick.

■ BIG NUMBERS: K, M, B, T

K is a thousand, M a million, B a billion, T a trillion, each a thousand times the last. Multiply the units as if they were numbers: K times K is M, M times K is B, M times M is T. Division runs the same way: M over K is K, and M over M leaves a plain number. So 2M e-bikes times \$2,000: 2 times 2, M times K, \$4B.

Capex of \$90M for the assembly plant over a net benefit of \$30M per year: M over M cancels, 90 over 30 is 3, so payback is 3 years. Breakeven volume: \$30M of fixed cost over the \$150 per unit the client saves. Move the decimal until the divisor

turns friendly: 30M over 150 is 3M over 15, which is 0.2M, which is 200K units per year. Say each decimal move out loud so Player 2 can follow the zeros.

■ PERCENT TRICKS

- 10% then scale. Move the decimal one place to the left, then multiply. 30% of the 2.7M people on the Wasatch Front: 10% is 270K, times 3 is 810K.
- 15% as 10% plus half. 15% of 1.2M Utah households: 120K plus 60K is 180K.
- 25% is a quarter and 12.5% an eighth. Reach for the scroll.
- Rule of 72. Divide 72 by the annual growth rate to get the doubling time. A market growing 8% a year doubles in about 9 years.
- Compounding creeps. For a few years at a modest rate, multiply rate by years, then add about a tenth of that product: three years at 10% is 30% plus 3%, so 33%, since each year grows on the prior year's gain.

■ FRIENDLY NUMBERS, SAID OUT LOUD

Round before you compute, and announce the rounding. Germany holds 84M people. Say "I will round to 80M for clean division, which understates by about 5%," then divide. Player 2 hears you control the error. Pick 5 or 8 instead of 7 for your own assumptions. If Player 2 handed you the 7, keep it. Round it to 8 only where the 7 multiplies, and say the answer runs about 14% high.

Then sense-check before you speak the answer:

- Units. Confirm the answer is a flow, dollars or units per year.
- Magnitude. Recount the zeros against a number you own. A \$33.6B Utah e-bike market would mean close to \$10,000 per resident per year, and you catch the slip from M to B.
- Round trip. Reverse the last step. If 400K times \$150 gave you \$60M, check that \$60M over 400K returns \$150.

■ CASTING OUT LOUD

Write the equation before you touch a number: "Households, times ownership rate, divided by replacement years, times price." Then look up and get the nod. Player 2 flags a missing step there for the price of ten seconds. Then narrate each step: "40M cyclists, 60% casual, so 24M." One coach said the failure mode is solving in silence for five minutes and then surfacing with a number.

Most consulting firms do not allow a calculator, Capital One does, and a former Bain consultant told the room that BCG runs video rounds with your hands and paper on camera, so ask the recruiter what your round allows. Build the habit for the firms that make you finish the arithmetic.

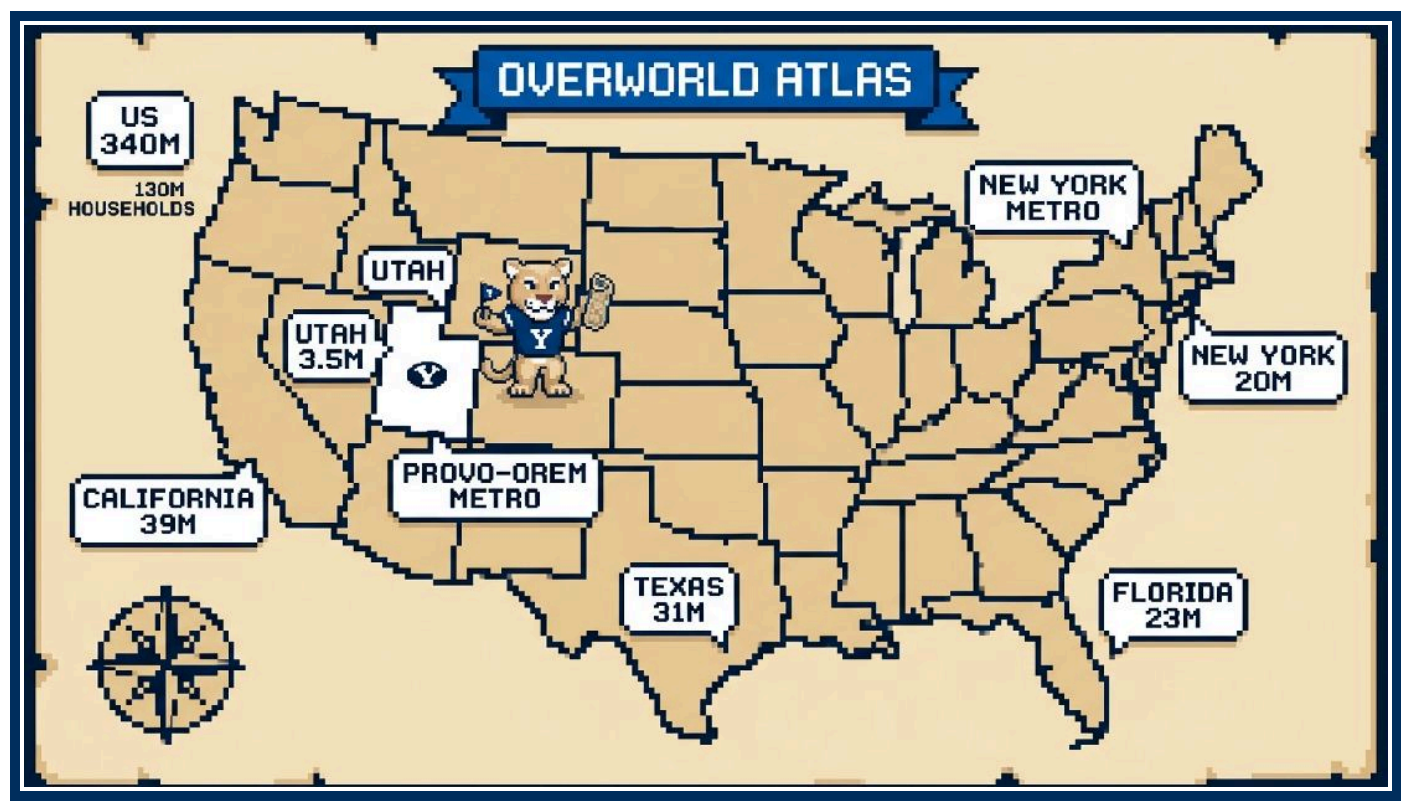
■ THE TEN-MINUTE DAILY DRILL

Run it out loud on a timer, with a partner at least twice a week.

THE TEN-MINUTE DAILY DRILL	
MINUTES	DRILL
0 to 2	Fraction Scroll flash: partner calls the fraction, you call the percent, then swap
2 to 4	Ten two-digit multiplications, spoken, no paper
4 to 6	Five percents of Overworld Atlas numbers, 10% then scale
6 to 8	Five divisions with the rounding announced, then the error estimate
8 to 10	One Sector Codex story problem: equation, number, one line of Loot

Log the misses, and after two weeks you can see which spell runs short on MP.

04 PLAYER INVENTORY II: THE OVERWORLD ATLAS



A Sizing Dungeon run and most Number Cruncher fights open on a population, a household count, or an age cohort. Player 2 hands you some of those figures and expects you to carry the rest. This atlas is the loadout.

■ THE NEED-TO-KNOW RULE

The coaches' short list is the minimum: the US population, the city you live in, the city you are interviewing in, two countries or one continent, and a sector's headline figure if the firm works in one sector (Medicare enrollees, 68M, for a healthcare boutique). For you, Cosmo, that means 340M for the US, Provo at 115K inside a Provo-Orem metro of 780K, the interview city (Dallas at 1.3M inside a Dallas-Fort Worth metro of 8M), and, since Wasatch Wheels sells in 25 countries, Germany at 84M, Canada at 40M, and Europe at 745M.

■ DERIVING HOUSEHOLDS

E-bikes can go either way, household or person, so say which base you chose: the Utah walkthrough in the Sizing Dungeon uses households, and the German walkthrough in Boss Fight I uses riders. Divide population by household size. The US runs 340M at 2.5 per household, and that division overshoots the atlas row of 130M, because 8M Americans live in dorms and care homes and the true average runs above 2.5. Use 130M. Utah runs 3.5M at 3.0, so 1.2M households. A former McKinsey engagement manager put Manhattan near 1.5. The Census

Bureau counts it nearer 2. Either passes the reasonable test if you say which one you are using and why. For a metro with no household row, borrow the state figure: Salt Lake City metro 1.3M divided by 3.0 is about 430K households.

■ DERIVING AGE COHORTS

Run the Spellbook's 1/80 trick. Spread an 80-year lifespan into even slices, and each year of age is 1.25% of the population, 4.2M people in the US. A cohort's share is its span in years times 1.25%, which lands within two points of the atlas rows, so use the trick for slices with no atlas row: ages 25 to 44 span 20 years, and 20 times 4.2M is about 84M Americans. Then tilt for the place, Utah young and Florida old, and say the tilt out loud so Player 2 can nod.

■ THE ATLAS

The metro is the market for most sizing, so say which figure you mean: New York City at 8.3M and the New York metro at 20M are different quests. Match the base to the client's channel: a dealer network sells to a metro, a state law applies to a state, and a bike-share contract with a city hall fits the city-proper figure.

UTAH	
FIGURE	VALUE
Utah population	3.5M
Utah households	1.2M
Utah average household size	3.0
Wasatch Front population	2.7M
Salt Lake County	1.2M
Utah County	720K
Salt Lake City (city proper)	210K
Salt Lake City metro	1.3M
Provo (city proper)	115K
Provo-Orem metro	780K
BYU total enrollment	34K



Top 10 States by Population	
State	Population
Michigan	10M
Washington	8M
Arizona	7.5M
Massachusetts	7M
Colorado	6M
Utah	3.5M
Nevada	3.3M
Idaho	2M

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METRO AREA	METRO POPULATION	CITY PROPER
New York metro	20M	8.3M
Los Angeles metro	13M	3.8M
Chicago metro	9.5M	2.7M
Dallas-Fort Worth metro	8M	1.3M (Dallas)
Houston metro	7.5M	2.3M
San Francisco Bay Area	7.5M	800K (San Francisco)
Washington DC metro	6.3M	700K
Atlanta metro	6.3M	500K
Miami metro	6.2M	n/a
Philadelphia metro	6.2M	1.55M
Phoenix metro	5M	1.7M
Boston metro	5M	650K
Seattle metro	4M	750K
Minneapolis-St. Paul metro	3.7M	n/a
Denver metro	3M	720K
Austin metro	2.5M	1M
Salt Lake City metro	1.3M	210K



COUNTRIES			
COUNTRY	POPULATION	COUNTRY	POPULATION
India	1.44B	Vietnam	100M
China	1.41B	Iran	90M
United States	340M	Turkey	86M
Indonesia	280M	Germany	84M
Pakistan	255M	United Kingdom	68M
Nigeria	230M	France	68M
Brazil	215M	Italy	59M
Bangladesh	175M	South Korea	52M
Russia	145M	Spain	48M
Mexico	130M	Canada	40M
Japan	124M	Saudi Arabia	36M
Philippines	115M	Australia	28M
Egypt	110M		

REGIONS	
REGION	POPULATION
Asia	4.8B
Africa	1.5B
Europe	745M
European Union	450M
Latin America and Caribbean	660M
North America (US, Canada, Mexico)	510M
South America	440M
Oceania	45M

■ THREE WORKED DERIVATIONS

State the assumptions first, then the arithmetic.

Utah households with a child under 18. Start with 3.5M people. The US under-18 share is 73M of 340M, about 21%. Utah households run 3.0 against the US 2.5, so read that as more families and tilt to 25%: about 875K children. Assume two children per family that has any, and you get about 440K households, more than a third of Utah's 1.2M. Loot: a cargo-bike push in Utah targets about 440K family households, and children per family is the assumption to firm up with a census pull before the model reaches Dana Okafor.

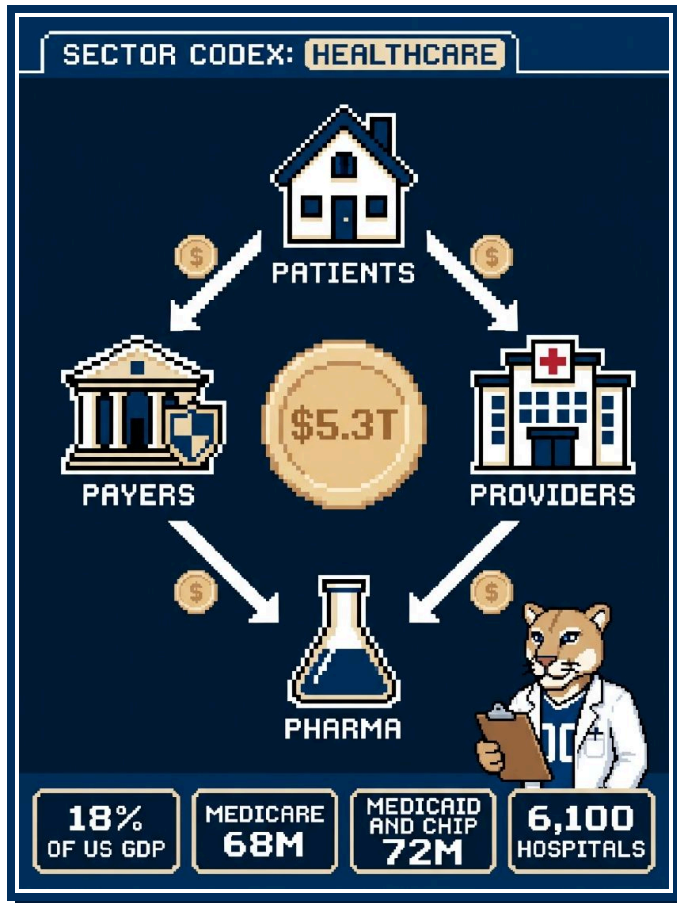
Workers along the Wasatch Front. The US labor force is 170M of 340M, one half, so assume 50%. Utah's extra children pull that share down, and its adults work at a higher rate than the country's, so hold 50% and say why. The Wasatch Front has 2.7M people, and 2.7M times 50% is about 1.35M workers. Loot: the commuter line's home market is about 1.35M workers, and the next quest is the share within a rideable commute.

Floridians 65 and over. Florida has 23M people. The national share is 60M of 340M, about 18%, and retirees move to Florida, so tilt to 20%. Then 23M times 20% is about 4.6M. Loot: one state holds close to 8% of the country's 65+ population, and the next quest is the share still riding, since ownership falls with age.

■ THE MISSING NUMBER

Ask. The coaches were blunt: Player 2 runs the case to watch what you do with a number and will hand you Argentina's population on request. If Player 2 returns the question with "what would you assume," anchor on a country of similar size that you do know: Argentina sits near Spain's 48M and Canada's 40M, so call it 45M, flag it as an assumption, and move on. If you reach Boss Fight I and find a figure from the Opening Cutscene unconfirmed, ask then. A wrong input poisons the math and the loot behind it.

05 PLAYER INVENTORY III: SECTOR CODEX, HEALTHCARE



A former McKinsey engagement manager told the room that a healthcare-focused firm expects you to know the Medicare enrollee count the way you know the US population.

■ FOLLOW THE MONEY

Four parties share \$5.3T a year, \$15.5K per person. Follow the money between them.

Patients receive care and pay a slice through premiums, deductibles, and copays. A payer sets or negotiates the price, so patient price sensitivity matters less than it does for e-bikes.

Payers collect premiums or tax dollars and pay claims: Medicare (68M: 65 and over, plus younger people with disabilities), Medicaid and CHIP (72M), employer plans (155M lives), and the ACA marketplace (24M), with 8% of people uninsured. Low-income seniors sit in two payer groups, so the figures do not sum to 340M.

Providers deliver care and bill the payer: 6,100 hospitals with 920K beds, 1M active physicians, 3.3M registered nurses. Hospital care takes 30% of spending, yet a typical hospital clears a 1% to 4% operating margin.

Pharma sells through three big wholesalers, and pharmacy benefit managers negotiate rebates for payers. Device makers sell to hospitals through group purchasing contracts, distributors, and their own reps. Generics fill 90% of prescriptions for 20% of drug dollars, so brands take 80% of the money. A new drug costs \$2B+, and 10% of Phase I entrants reach approval.

A prompt about who pays whom (a payment model, a pharma launch, a payer entering a market) splits into these four regions. A profitability or entry prompt follows its own skill tree, with the payer flow inside the revenue region.

■ THE CODEX TABLE

ENTRY	FIGURE
US healthcare spending	\$5.3T
Healthcare share of US GDP	18%
Spending per person	\$15.5K
Hospital care share of spending	30%
Physician share of spending	20%
Retail drugs share of spending	9%
Medicare enrollees	68M
Medicaid and CHIP enrollees	72M
Employer-sponsored covered lives	155M
ACA marketplace enrollees	24M
Uninsured share of population	8%
US hospitals	6,100
US hospital beds	920K
Active physicians	1M
Registered nurses	3.3M
Typical hospital operating margin	1% to 4%
Average hospital length of stay	4.5 days
Typical hospital occupancy	65% to 70%
Medicare 30-day readmission rate	15%
MLR floor, individual and small group	80%
MLR floor, large group	85%
Generics share of prescriptions	90%
Generics share of drug spend	20%
Cost of a new drug to market	\$2B+
Drug development timeline	10 to 15 years
Phase I to approval success rate	10%

■ METRICS IN CASES

- **PMPM** (per member per month): a plan's revenue or medical cost per covered life per month, so members times 12 times PMPM gives the year.
- **MLR** (medical loss ratio): the share of premium dollars an insurer pays out as claims, with legal floors of 80% and 85%.
- **ALOS** (average length of stay): days per inpatient admission, 4.5 on average. Under fixed per-admission payment, a shorter stay earns the same revenue at lower cost.
- **Occupancy**: occupied over staffed beds, 65% to 70% for a typical hospital. Beds times occupancy times 365 gives bed-days, and bed-days divided by ALOS gives admissions.
- **Readmissions**: discharged patients back inside 30 days, 15% for Medicare, which fines hospitals above their expected rate.
- **Payer mix**: a provider's revenue split across Medicare, Medicaid, commercial, and self-pay. Commercial pays the most, so a richer commercial mix lifts margin without touching volume.
- **Case mix**: an index of patient sickness. Sicker patients cost and pay more per admission, so compare margins across similar case mix.

■ TWO MINI PROMPTS

Prompt one, sized. Player 2 says: a 300-bed Wasatch Front hospital wants 15% more inpatient revenue without building. Size current capacity and find the lever.

Run it as a Sizing Dungeon and flag \$15K per admission as your least-supported number.

FLOOR	ASSUMPTION	CALCULATION
Staffed beds	prompt	300
Occupied beds	70% occupancy	300 times 70% is 210
Bed-days	365 days	210 times 365 is about 77K
Admissions	ALOS 4.5 days	77K divided by 4.5 is about 17K
Revenue	\$15K per admission	17K times \$15K is about \$255M

Revenue is admissions times revenue per admission, so split the levers into volume (ALOS, occupancy) and rate (contract renewal, payer mix, case mix). Start with volume: the prompt rules out building, and a rate move waits on the next contract cycle.

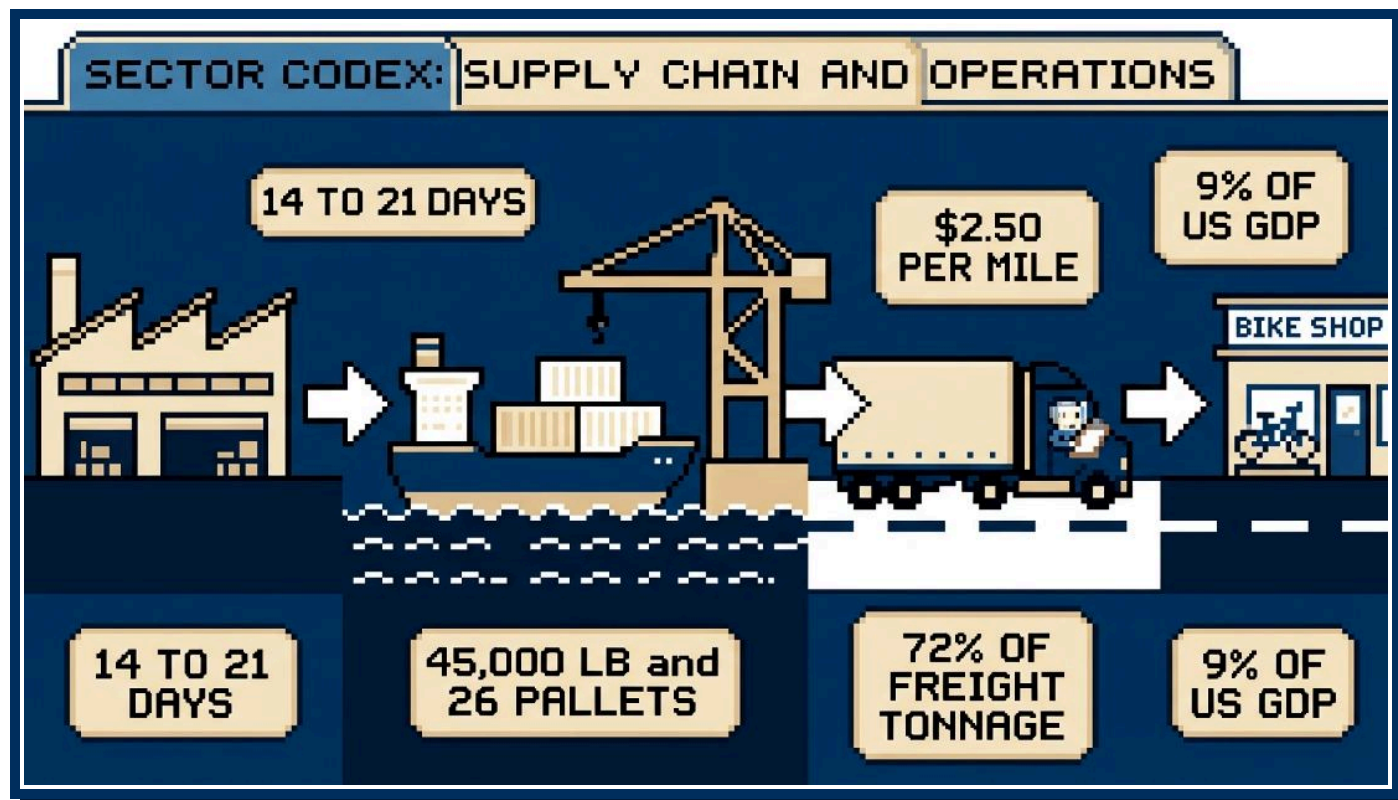
The Loot: ALOS from 4.5 to 4.0 days adds 12.5% more admissions, and occupancy from 70% to 80% adds 14% more bed-days, so neither move reaches 15% on its own. ALOS to 3.9 days does (4.5 divided by 3.9 is about 1.15). Pull neither lever until you confirm the hospital turns patients away today: check ED boarding hours and transfer refusals. Lead with ALOS, since a shorter stay costs less per admission and more filled beds mean more nurses. Next: ALOS by service line and post-acute discharge bottlenecks.

Prompt two, structured. Player 2 says: a regional health plan with 500K members may subsidize a Wasatch Wheels commuter e-bike for members who ride to work. Structure whether it pays.

Build a Quest Map with four regions. Cost: subsidy times uptake (2% of 500K is 10K bikes at \$500, so \$5M), admin, and injury claims. Benefit: the medical cost PMPM each rider avoids, times riders, times months, plus the members the perk keeps. Delivery: dealer fulfillment, eligibility rules, and fraud checks. Risk: selection bias, since members who claim the subsidy already ride and bring no new savings, and churn of 15% to 20% a year.

Run the breakeven before Player 2 asks: \$5M across 10K riders over 12 months needs \$42 PMPM per rider, a 7% cut of commercial medical cost near \$600 PMPM in one year from one bike, which fails the reasonable test. Hold the one-year test, because churn takes the rider before a longer payback lands. The Loot: the medical savings fall short on their own, so run the program as a retention play or point it at a high-cost cohort. First analysis: claims for today's bike commuters against matched members who do not ride.

06 PLAYER INVENTORY III: SECTOR CODEX, SUPPLY CHAIN AND OPERATIONS



Operations cases carry more numbers per minute than any other boss form, so load this page into MP first. Logistics costs the US about 9% of GDP, \$2.5T a year, and trucks carry 72% of freight tonnage.

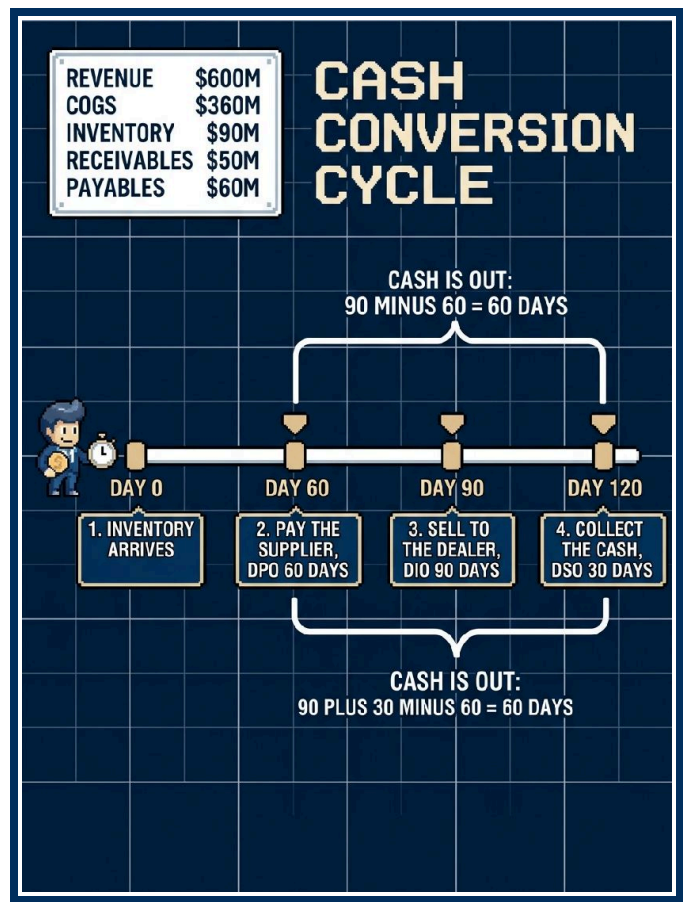
Wasatch Wheels has revenue of \$600M. Assume cost of goods sold \$360M, inventory \$90M, dealer receivables \$50M, and supplier payables \$60M.

■ INVENTORY TURNS

Inventory turns equal cost of goods sold divided by average inventory: how often a year the warehouse empties and refills. Benchmarks: grocery 12 to 15, general merchandise retail 4 to 8, industrial manufacturing 4 to 6, auto dealers 3 to 5. Holding stock costs 20% to 30% of its value each year in capital, space, and obsolescence, so a client turning slower than its sector has cash asleep on the shelf.

■ DIO, DSO, DPO, AND THE CASH CONVERSION CYCLE

Days inventory outstanding (DIO), days sales outstanding (DSO), and days payables outstanding (DPO) divide inventory, receivables, and payables by their daily flows: cost of goods sold for DIO and DPO, revenue for DSO. DIO equals 365 divided by turns. The cash conversion cycle is DIO plus DSO minus DPO: the days you fund the company between paying suppliers and collecting from dealers.



- Turns: $\$360\text{M}$ divided by $\$90\text{M} = 4$, so $\text{DIO} = 365$ divided by 4, about 90 days
- DSO: $\$50\text{M}$ divided by $\$600\text{M}$, times 365, about 30 days
- DPO: $\$60\text{M}$ divided by $\$360\text{M}$, times 365, about 60 days
- Cash conversion cycle: $90 \text{ plus } 30 \text{ minus } 60 = 60 \text{ days}$

The Loot: one day of inventory ties up about $\$1\text{M}$ ($\$360\text{M}$ divided by 365), so cutting DIO by 10 days frees $\$10\text{M}$ and saves $\$2\text{M}$ to $\$3\text{M}$ a year in carrying cost.

■ SERVICE LEVELS

Fill rate is the share of demand you ship from stock on the first try. Target 95% to 98%. OTIF means on time and in full against the requested date. Typical is 85% to 95%, best-in-class 95%+. Perfect order adds damage free and billed clean. Multiply the four rates: 95% times 97% times 99% times 98% is 89%, a surprise to executives holding four high scores. Inventory accuracy, system count against shelf count, belongs at 98%+. Below that, you promise dealers stock your pickers cannot find.

■ STOCK RULES: SAFETY STOCK, EOQ, ABC

Safety stock is the buffer above forecast for demand above plan and replenishment behind schedule. It grows with your service level target, demand variability, and lead-time length and

variability, so a two- to three-week ocean leg from Taiwan needs more buffer than a truck from Utah.

Economic order quantity balances ordering cost against carrying cost. The ideal order grows with the square root of demand, so doubling volume raises it by about 40%.

ABC classification sorts SKUs by value: 20% of SKUs drive 80% of value. Count A items each week and run C items on reorder points.

■ THE FACTORY: OEE, TAKT, CYCLE, THROUGHPUT, LITTLE'S LAW

Overall equipment effectiveness multiplies availability (uptime over scheduled time), performance (actual over rated speed), and quality (first-pass yield). Typical plants run 60%, and world class is 85%. US manufacturing capacity utilization sits near 77%, so most plants have room before capex.

Takt time equals available production time divided by demand. Wasatch Wheels builds 400K bikes a year on 250 days of two 8-hour shifts, 240K minutes, so takt is 240K divided by 400K: one bike per 36 seconds, 100 an hour. Cycle time is one station's time per unit. The slowest station is the bottleneck, and throughput, units completed per hour, drops to its pace.

Little's Law: work in process equals throughput times flow time. At 100 bikes an hour and three hours from frame to box, the floor holds 300 bikes. Cut flow time to two hours and 100 bikes leave the floor, about $\$90\text{K}$ at $\$900$ of cost each.

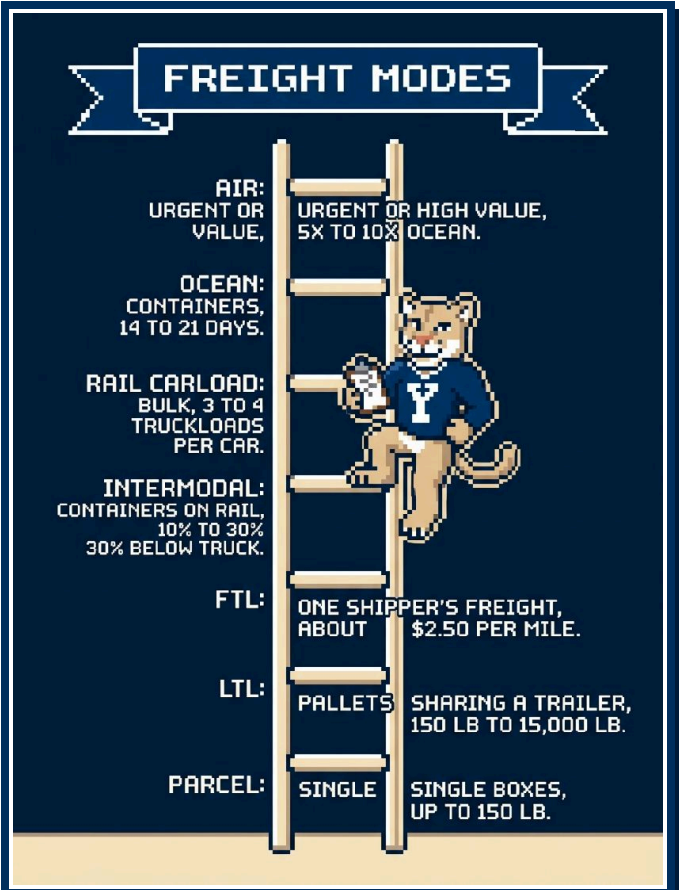


■ THE WAREHOUSE

Pick rate counts lines per labor hour: 100 to 200 picks per hour by hand, 300+ picks per hour semi-automated (pick-to-light, conveyors), and goods-to-person systems run higher. Space

utilization should run 80% to 85%, or aisles clog and picks slow. Dock-to-stock, trailer arrival to pickable stock, should run under 24 hours. Cost per order equals warehouse cost divided by orders you ship, and warehousing runs 2% to 5% of sales: for Wasatch Wheels, 3% of \$600M is \$18M, about \$45 per bike, the bar for any automation proposal.

■ FREIGHT MODES



MODE	CARGO	CAPACITY	COST LOGIC
Parcel	Single boxes	Up to 150 lb per package	By weight and zone. You pay dimensional weight on bulky boxes
LTL	Pallets sharing a trailer	150 lb to 15,000 lb per shipment	By weight, class, and distance. Hub handling adds days
FTL	One shipper's freight	53 ft dry van, 45,000 lb, 26 pallets	About \$2.50 per mile all-in, so fill the truck
Intermodal	Containers on rail, truck at each end	One container, about a truckload	10% to 30% below truck on 700+ mile lanes, and slower
Rail carload	Bulk: grain, coal, chemicals	Three to four truckloads per car	Cheapest per ton-mile on land
Ocean	Containers	40-foot box: 2,390 cu ft, 58,000 lb. Largest ships carry 24,000 TEU	Per container, so density sets unit cost. Shanghai to Los Angeles: 14 to 21 days, \$2,000 to \$3,000 per box in a normal market
Air	Urgent or high value	Belly holds on passenger flights or freighters, ULD pallets and containers	Chargeable weight, the greater of actual and volumetric. 5x to 10x ocean per kg. You buy days

Bikes fill a container by volume long before its weight limit.

■ DRAYAGE AND THE LAST MILE

Drayage is the short truck move from port to rail ramp or warehouse. Carriers price it per container plus detention (you hold the box at your dock past free days) and demurrage (it sits at the terminal past free days). Chassis rental is a third line. Los Angeles and Long Beach handle 30% of US containerized imports, so a backup there delays dealers in Ohio. The last mile eats 40% to 50% of the cost to deliver a package, because a driver's hour buys a fixed number of stops. Route density is the lever.

■ LANDED COST AND INCOTERMS

Landed cost is the full cost of a unit at your door, the number to compare suppliers on: factory price, freight, insurance, duties, drayage, handling, and carrying cost while afloat. Incoterms set who pays each leg and where risk changes hands. FOB: the seller loads the vessel, and the buyer owns the ocean leg onward. DDP: the seller delivers to your door with duties paid, so ask which tariff rate sits inside the quote.

■ THE BULLWHIP

Small wobbles in end demand grow into large swings upstream because each buyer up the chain reads its customer's orders as the market, batches orders, pads safety stock, and over-orders in shortages. Fixes: share point-of-sale data, order smaller lots more often, cut promotions, and let vendors manage customer stock. At Wasatch Wheels, 1,200 dealers each padding a spring order can leave you a fifth overbuilt and discounting in September.

■ NEARSHORING AND TARIFFS

Nearshoring moves production from Asia to Mexico, and reshoring brings it home to the US. Both shorten lead time, cut tariff exposure, and shrink safety stock. You trade higher labor cost for lower freight, duties, and cash in transit. Compare on

landed cost per unit plus the working-capital release, and state your tariff assumption, because policy moves faster than a plant. A sourcing program saves 5% to 15% of addressed spend.

■ THE DRIVER SHORTAGE

The US runs 60K to 80K truck drivers short, and hours-of-service rules cap driving at 11 hours a day. Rates rise when capacity tightens, shippers move long lanes to intermodal, and carriers give first call to shippers who load fast and let drivers swap trailers. A 500-mile lane fits one driver-day. A 1,200-mile lane needs a relay, a team, or a train.

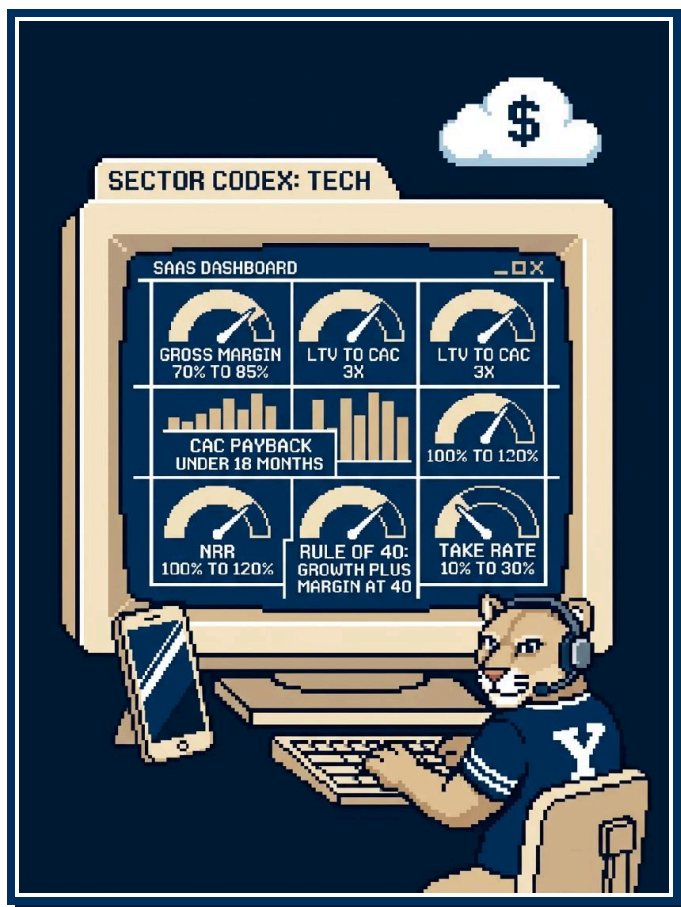
■ THREE MINI PROMPTS

Late and short. Player 2 says dealers report spring orders arriving late and short. Split OTIF in two. On-time misses point to transport: carrier acceptance, warehouse order-to-ship time, peak-season mode choice. In-full misses point to stock: forecast error by SKU, safety stock on A items, slow dock-to-stock on Taiwan containers. Ask for OTIF by month and region to learn which half is broken.

The ocean freight bill. Size Wasatch Wheels' annual ocean spend. All 400K bikes leave Taiwan by sea at 200 per container: 2,000 containers at \$2,500, about \$5M, under 1% of revenue, and the US leg is 70% of it. The Loot: ocean freight is a rounding error, and the cost conversation belongs to tariffs.

Free the cash. The CFO wants \$20M out of working capital. Build the Quest Map on the cash conversion cycle. Region one, inventory days: rationalize the C tail, rebuild safety stock on fresh forecasts, shorten lead time through nearshoring. At \$1M of cost of goods a day, \$20M from this region alone is 20 days of DIO, 90 to 70. Region two, dealer receivables: tighten terms or offer an early-pay discount. Region three, supplier payables: extend terms without breaking the relationship. Quantify each region in days, then start with the A items.

07 PLAYER INVENTORY III: SECTOR CODEX, TECH AND BONUS PAGES



Tech cases turn up at firms with no tech practice, because a soap maker and a police department both carry an app and a cloud bill now. The benchmarks below are your reasonable test.

■ SAAS UNIT ECONOMICS

Revenue. MRR is monthly recurring revenue: paying subscribers times average monthly price. ARR is MRR times 12. Setup fees and services stay out of both.

Retention. Gross churn is the share of customers, or of revenue, you lose in a period before counting upsells. Monthly churn compounds: 3% a month removes about 30% of the base in a year. Net revenue retention (NRR) is this year's revenue from last year's customers divided by what they paid last year, after churn, downgrades, and upsells. An NRR above 100% means the base grows before sales signs a new logo. Name the measure before you quote a churn figure.

Acquisition. CAC is sales and marketing spend divided by the new customers you won in the period. LTV is the gross profit you earn from one customer over its life: annual revenue per customer times gross margin, divided by annual churn. One

over churn is customer life in years, so 20% churn means five years. CAC payback is CAC divided by monthly gross profit per customer.

Margin and efficiency. Software gross margin runs high because one more user costs pennies to serve. Hardware and marketplaces carry cost of goods or seller payouts. Rule of 40: add revenue growth percent to profit margin percent. The magic number is the quarter's increase in quarterly recurring revenue, times four, divided by the prior quarter's sales and marketing spend. The burn multiple is net cash burned divided by net new ARR, and a lower number means cheaper growth.

Engagement and funnel. Divide daily active users by monthly active users to get DAU/MAU, your stickiness gauge. Chain the funnel conversions: visitors to trials, trials to paid, or free users to paid under freemium.

Take rates. A take rate is the platform's cut of each transaction. Marketplaces, app stores, and payment processors each charge one, and Mini Prompt 2 below turns on the gap between two of them.

TECH BENCHMARKS

METRIC	BENCHMARK
Software gross margin	70% to 85%
Hardware or marketplace gross margin	40% to 60%
LTV to CAC	3x
CAC payback	under 18 months
NRR, good	100% to 120%
NRR, best in class	120%+
Monthly gross churn, SMB	3% to 5%
Monthly gross churn, enterprise	under 1%
Annual gross churn, healthy SaaS	5% to 10%
Rule of 40	growth % + profit margin % $\geq$ 40
Magic number	0.75+
Burn multiple, good	under 1.5
DAU/MAU, good	20%+
DAU/MAU, social apps	50%+
Visitor to trial conversion	2% to 5%
Trial to paid conversion	10% to 25%
Freemium to paid conversion	2% to 5%
Marketplace take rate	10% to 30%
App store take rate	15% to 30%
Payments take rate	3%
SaaS R&D share of revenue	20% to 30%
Sales and marketing share, growth stage	30% to 50%
Public SaaS EV to revenue	5x to 10x

Market figures: global IT spending \$6T; cloud share AWS 28%, Azure 21%, Google Cloud 14%; global internet users 6B and global smartphone users 4.9B; US smartphone penetration 90% and US e-commerce share of retail 16%.

MINI PROMPT 1: THE CONNECTED-BIKE SUBSCRIPTION

Player 2 reads: Wasatch Wheels wants to sell a \$10 per month app subscription (theft tracking, battery health) to new bike buyers. Building the app costs \$3M. Dana wants to know whether it is worth building.

Set up the equation and get the nod before you touch a number. Subscribers equal units sold times attach rate. ARR equals subscribers times \$120. LTV equals \$120 times gross

margin divided by annual churn. Then compare LTV to CAC, compute payback, and compare first-cohort gross profit to the build cost.

Solve out loud. Assume 20% of 400K buyers attach: 80K subscribers and \$9.6M of new ARR per cohort. Assume 80% gross margin and 20% annual churn, so LTV is \$120 times 80% times 5, or \$480. CAC is a \$40 dealer incentive per activation, so LTV to CAC is 12x and payback is five months (\$40 divided by \$8 of monthly gross profit). Against a \$3M build, the first cohort's \$7.7M of gross profit (\$9.6M times 80%) covers \$3.2M of dealer incentives and pays back the build inside a year.

The Loot: an 80% margin software line on a hardware company is worth building. The least supported number is the 20% attach rate. Next step: pilot at 50 dealers and measure attach and 90-day churn.

MINI PROMPT 2: THE MARKETPLACE OFFER

Player 2 reads: A national online marketplace offers to list Wasatch Wheels bikes at a 15% take rate and claims it will add 20K units a year. The website sells 40K units today and pays a 3% payments take rate. Dana wants a recommendation.

Structure it as three quests. Fee math: 20K units times \$2,000 is \$40M of gross sales, and 15% of that is \$6M in fees, against \$1.2M if the same 20K units ran through the website at the 3% payments rate. That \$4.8M gap shrinks once you count the website's own marketing spend and the margin a dealer keeps on a bike sold in a shop. Cannibalization: the fee earns its keep only on units that would otherwise go unsold, so find the share of the 20K that shifts from the website or the 1,200 dealers. Channel conflict: dealers asking for price support will read a marketplace listing as a second attack on their margin.

The Loot: you pay the \$4.8M fee gap for growth only if most of the 20K units are new demand. Next step: get new-to-brand buyer data from the marketplace and run a one-region test.

BONUS CODEX

Retail and CPG. Grocery gross margin is 25% to 30% and grocery net margin is 1% to 3%, so cut price by one point and you erase a third of net profit, or all of it. Apparel gross margin runs 50% to 60%. Retail sales per square foot (typical) sit at \$300 to \$600. In-store conversion runs 20% to 40% against e-commerce conversion of 2% to 3%. Retail shrink is 1.5% to 2% of sales. CPG trade spend share of revenue is 15% to 25%.

Financial Services. Bank net interest margin is 3%: interest earned minus interest paid, divided by earning assets. Bank efficiency ratio, non-interest expense over revenue, is 55% to 60%, and lower wins. Bank ROE is 10% to 15%. Credit card interchange is 2%, and merchants pay it to the card issuer.

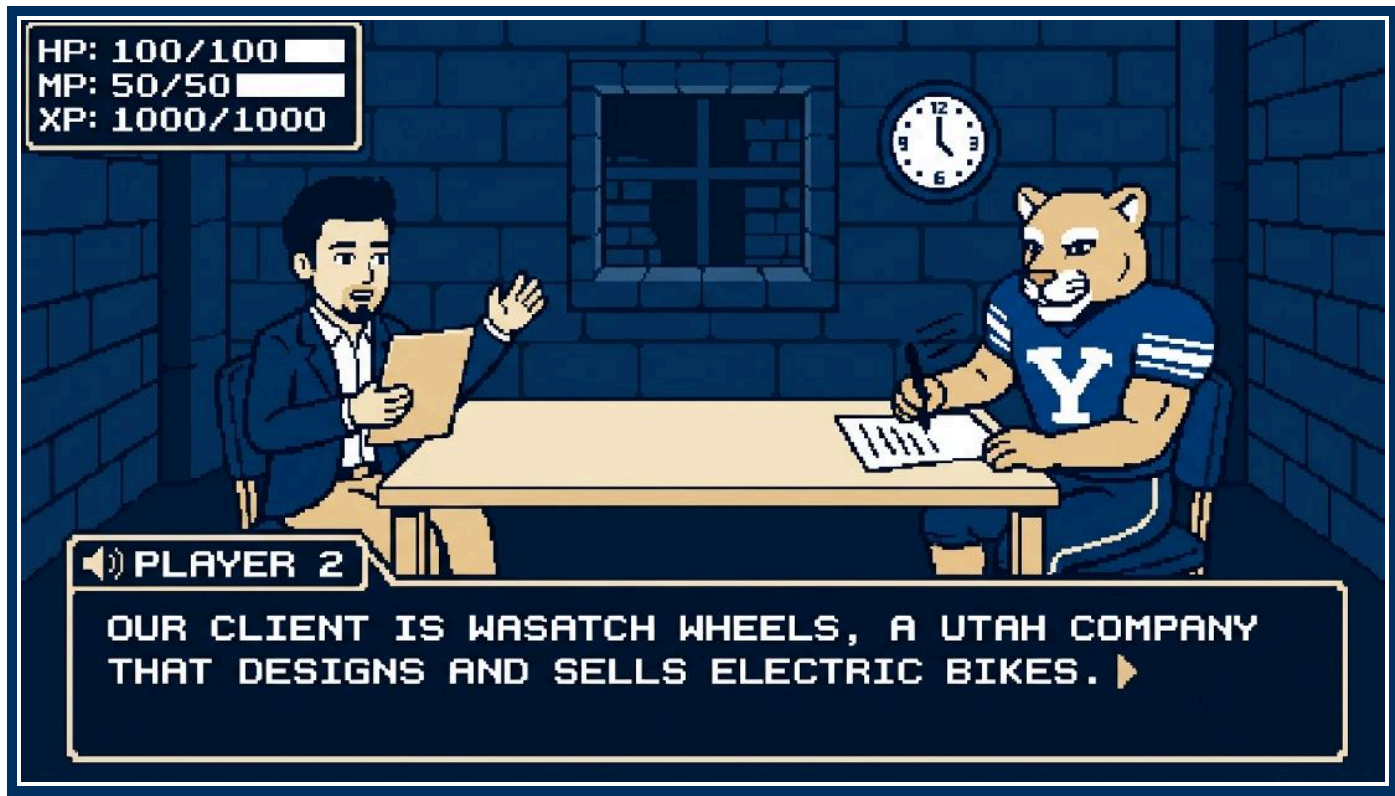
In a Capital One case you build the P&L from interchange plus interest income, minus credit losses and acquisition cost per account.

Private Equity. Mid-market EV/EBITDA multiple is 8x to 12x and LBO leverage is 4x to 6x EBITDA. A sponsor keeps the company for a PE hold period of 3 to 7 years and underwrites to a PE target IRR of 20% to 25% and a PE target MOIC of 2x to 3x. In a PE case you test three things: EBITDA growth, whether the multiple holds at exit, and whether the company can pay the debt down.

Airlines. Airline load factor is 83% to 85%, the share of seat-miles sold. Airline fuel share of costs is 20% to 30% and airline labor share of costs is 30% to 35%, so a fuel spike or a pilot contract moves the whole P&L.

Restaurants. Restaurant food cost is 28% to 35% of sales and restaurant labor cost is 25% to 35%. Together they form prime cost, and the restaurant prime cost target is under 65%. Restaurant occupancy (rent) cost is 6% to 10%. In a same-store sales case you split traffic from average ticket, and you treat a menu price case as an elasticity case wearing an apron.

08 LEVEL 1: THE OPENING CUTSCENE



Player 2 reads the prompt, and the Boss Fight starts before you pick up a pen. You run the math on the numbers you capture now and build the Quest Map around the objective you confirm now. Your HP bar is full. Protect it by listening more than you talk.

■ SET UP THE PAGE

Turn one sheet sideways and divide it before Player 2 starts talking. Box the client and the objective in a strip across the top. Run facts down the left column, and keep a narrow lane along its edge where you circle each figure on its own line. Save the bottom strip for questions. Keep a blank second sheet for the Quest Map and a third for math.

Write fast and abbreviate. Prompts run long, and the coaches called note speed a trained skill. "WW, #2 NA e-bike, \$600M rev, 400K units, 25 ctry, 70% US" fits on two lines and holds five numbers. Write the objective in full, in the client's words, since you build the map from that sentence.

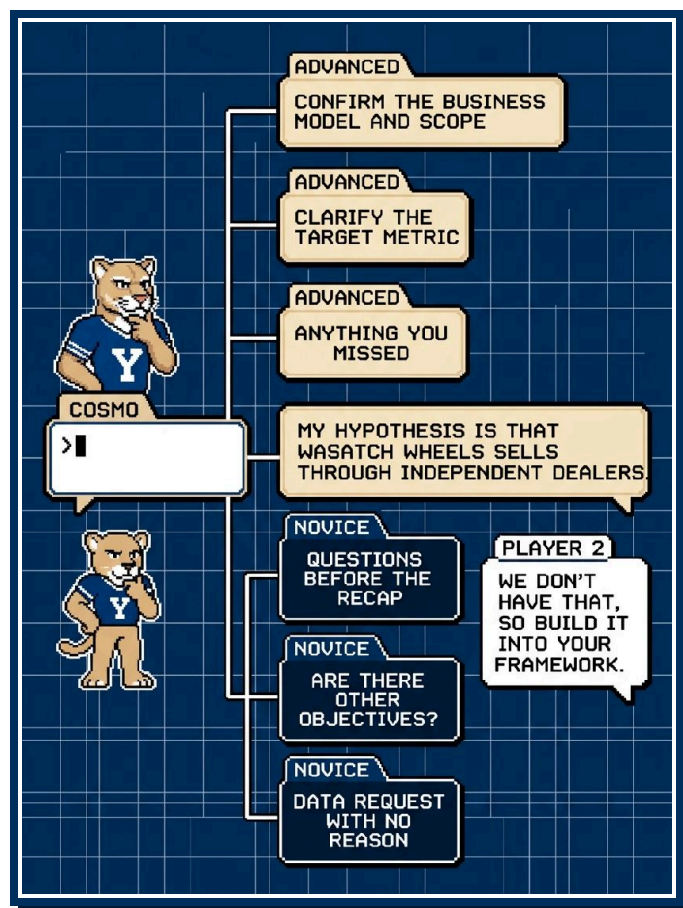
■ THE SAVE POINT

Recap out loud, in about one minute, in your own order, and confirm the numbers as you go. Over video, sixty and sixteen sound alike, so read each figure back with its unit and, where two numbers could blur, its digits: "six hundred million dollars, six-zero-zero," and "four hundred thousand units, not forty."

Categorize as you recap: company facts (founding, headcount, product lines), scale (revenue, units, rank), footprint (countries, US share), and the objective. A coach praised a student who recapped a math prompt by segment rather than in the order given, since the grouping kept the room organized once the math started.

End with the objective and a check: "So the goal is a growth strategy for Wasatch Wheels. Do I have that right?" Player 2 nods, and you have your save. Carry a wrong number past this point and you play the rest of the fight from a corrupted file.

THE DIALOGUE TREE



Ask three to five clarifying questions. You may ask none, though confirming a figure is cheap insurance. Player 2 answers advanced branches and dodges analysis in disguise, and a dodge says nothing bad about you. Player 2 also hears a novice branch for what it is.

Novice branches:

- Questions before the recap. Player 2 has no evidence yet that you heard the prompt.
- "Are there other objectives?" Player 2 stated the objective in the prompt, and asking again sounds like stalling.
- Data requests with no reason attached. Player 2 dodges "Do you have sales by region?" and marks you as fishing.

PLAYER 2

Our client is Wasatch Wheels, a Utah company that designs and sells electric bikes. Two former ski-lift mechanics founded it in Ogden in 2012 so they could ride to the resort without a car. It is the number two e-bike brand in North America by units, with \$600M in revenue, about 900 employees, and 400K units sold per year across three lines: commuter, recreation, and cargo. It sells in 25 countries, and 70% of revenue comes from the US. The CEO, Dana Okafor, has asked us to develop a growth strategy.

Advanced branches:

- Confirm the business model and the scope. "Does the growth question cover all three product lines and all 25 countries?"
- Clarify the target metric. "Does Dana have a revenue or unit target in mind, or a timeline?" Some case writers set no target, so a shrug here costs you nothing.
- Anything you missed. A number you failed to write down is fair game.

Phrase a question as a hypothesis and you hand Player 2 evidence of your thinking: "My hypothesis is that Wasatch Wheels sells through independent dealers. Do they also sell direct?" Attach a reason to each data request in the same spirit: "I'd like revenue by product line so that I can see which line is carrying the growth."

Player 2 phrases a dodge as "We don't have that data, so build it into your framework." The question belonged in the Quest Map, and you asked a good one. Note it in your questions strip and move on. Keep the phrase "dumb question" out of the room: the coaches watched strong candidates take a dodge and then talk themselves down for no reason.

You can rescue a number mid-case. If you reach the math and realize you skipped a figure, ask then. Ten seconds on the question beats a calculation built on a wrong number.

Bring courage to unfamiliar industries. If e-bikes and dealer sales are new to you, say so and ask. A former McKinsey engagement manager told the room he would rather spend a minute explaining a product category than staff an associate who nods along and learns nothing.

ASK FOR THE PAUSE MENU

Close the tree with one line: "I think I have what I need. I'd like to take two minutes to organize my thoughts, and then I'll walk you through my approach." Player 2 has watched hundreds of heads bend over paper and remembers none of them.

MOCK DIALOGUE: THE WASATCH WHEELS OPENING

[COSMO WRITES THROUGH THE WHOLE PROMPT AND CIRCLES EACH NUMBER]

COSMO

Let me play that back. Company facts: Wasatch Wheels, founded in Ogden in 2012 by two former ski-lift mechanics, about 900 employees, three lines: commuter, recreation, and cargo. Scale: \$600M in revenue, six hundred million dollars, six-zero-zero. I heard units as forty thousand a year, which would make the average bike \$15,000, so I assume four hundred thousand.

**PLAYER 2**

Four hundred thousand. 400K units, not 40K.

**COSMO**

400K, number two in North America by units. Footprint: 25 countries, 70% of revenue from the US. Objective: a growth strategy for the CEO, Dana Okafor. Do I have that right?

**PLAYER 2**

That matches.

**COSMO**

A few clarifying questions. First, on the objective: does Dana have a revenue or unit target in mind, or a timeline?

**PLAYER 2**

No target and no fixed timeline. She wants a plan she can take to her board.

**COSMO**

Second, on scope: are all three product lines and all 25 countries in play, or is she focused on the US?

**PLAYER 2**

Everything is in play.

**COSMO**

Third, on the business model. My hypothesis is that Wasatch Wheels sells through independent bike dealers. Do they also sell direct?

**PLAYER 2**

Both. About 1,200 independent dealers, plus their own website.

**COSMO**

Last one: do we have revenue split by product line? I'd like it so that I can see which line is carrying the growth.

**PLAYER 2**

We don't have that broken out. Include it in your framework.

**COSMO**

Understood, I'll build it in. I have what I need, so I'll take two minutes to organize my thoughts and come back with my approach.



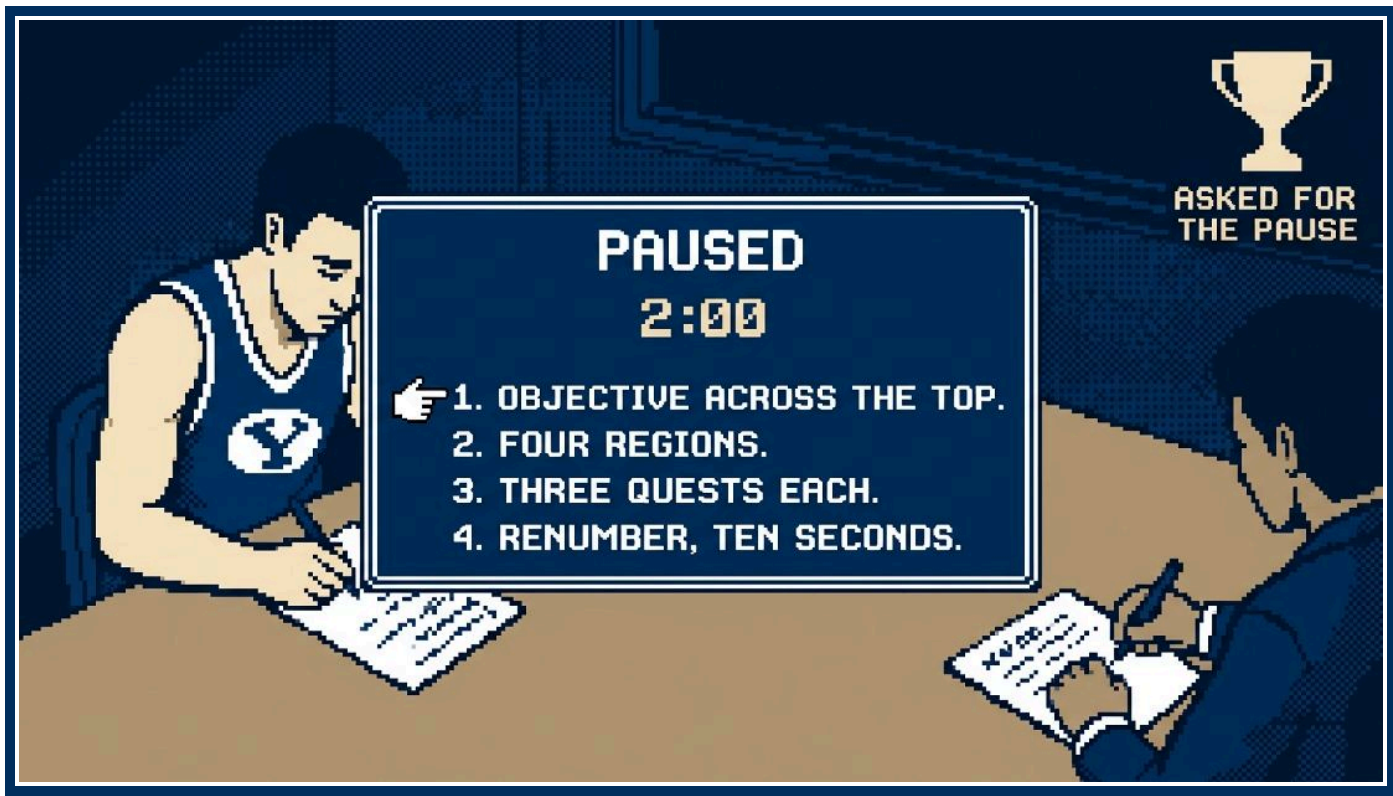
PLAYER 2

Take your time.



[COSMO TURNS TO THE BLANK SECOND SHEET AND STARTS THE QUEST MAP]

09 LEVEL 2: THE QUEST MAP



The Dialogue Tree closes, and you turn a two-paragraph prompt into a plan a team could run on day one.

■ THE PAUSE MENU

Ask for it: "Thanks, that helps. I'd like two or three minutes to organize my thoughts, and then I'll walk you through my approach." The former McKinsey engagement manager said any trained interviewer expects this request.

Use the two minutes in a fixed order.

- 1 Write the objective across the top, with the target metric if Player 2 gave you one. For Wasatch Wheels: "Growth, \$600M today, no target, assume +\$100M." \$100M is one sixth of \$600M, about 16.7%, and you label it as your own assumption when you deliver.
- 2 Draw four columns and label them in the order they come to mind.
- 3 Under each region, write three quests in shorthand. "Mkt size + growth by country/line" is enough on paper.
- 4 Spend the last ten seconds on the renumber: decide which region matters most for this client and write 1 through 4 beside the labels. Player 2 hears a ranked plan.

■ THE 4 BY 3 GRID

The grid holds four regions with three quests each. Four is the training default the coaches gave the room. Draw three and you fold two workstreams into one region and present them as one. Draw five or six and your walkthrough runs past the three-minute mark before you reach your compass. Three quests give a region enough weight to stand as a workstream.

A case is day one of a project, compressed. Three or four consultants land at the client with 4 to 12 weeks to answer a question the client could not, and the manager hands each of them a region from the whiteboard. The former McKinsey engagement manager described the failure mode: miss a region, and two weeks in you discover you need a consumer survey that takes two and a half weeks to design, field, and read.

■ NO OVERLAPPING REGIONS, NO UNEXPLORED MAP

Consultants call it MECE: mutually exclusive, collectively exhaustive. Each quest belongs in one region, and the four regions, taken as a set, cover the ways the client could reach the objective.

A bad map for Wasatch Wheels growth: Marketing, Customers, Dealers, Europe. Under Marketing and Customers you would run the same questions twice. Dealers is one channel out of two,

and Europe is one geography among 25 countries. Both are quests dressed up as regions. Line economics and the company's capacity to fund growth are missing, so the map has blank space where the money lives.

The fixed map: Market, Customer, Line Economics, Company and Channels. Test the exclusive half by naming the one region each quest belongs to, and merge or split when you can name two. Test the exhaustive half by finding a home for each growth lever: a price increase starts with willingness to pay in Customer, a new country in Market, a dealer acquisition in Company and Channels. A lever with no home means a missing region.

■ QUANTIFY EACH QUEST AND ATTACH A "SO THAT"

A quest is an analysis with a number in it and a reason to run it. "Look at customers" is a wish. "Share of revenue from the top three models, so that we know how much white space is left" is a quest an analyst can start on Monday. With the "so that," you tie the data you would pull to the client's decision. The former Bain consultant asked a student for the reason behind the customer quest, heard a general answer about understanding the buyer, and supplied the sharper version: demographics by

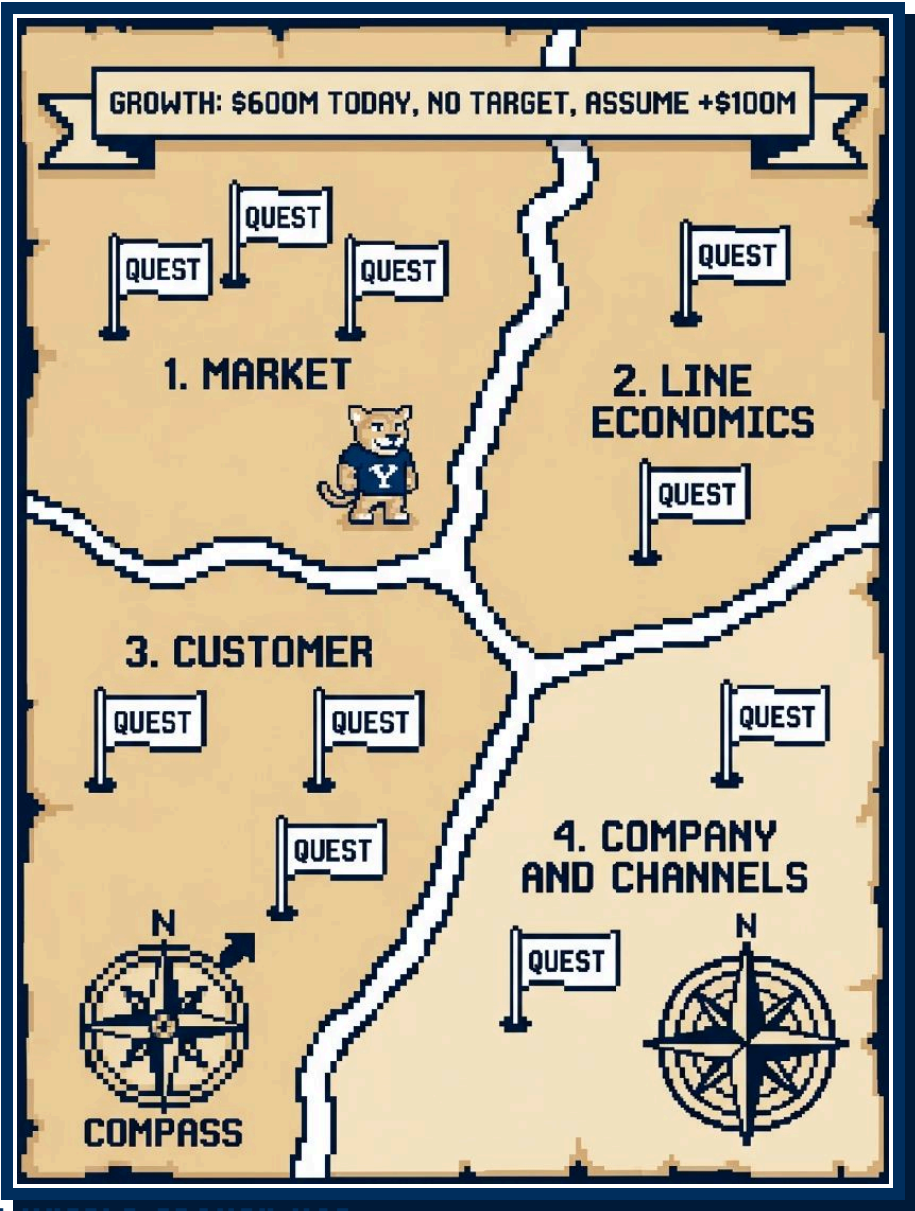
segment, so that we know which segments to push and which the brand leaves untouched. Write the sharper version the first time.

■ YOUR COMPASS

A hypothesis is the compass that points at the region to start in. State it at the end of the walkthrough, and use it during the pause, because the region it points at is the one you number 1. For Wasatch Wheels: "My compass points at the Market region. The US supplies 70% of revenue, and the biggest lever is whether the other 24 countries can carry the bulk of the \$100M we assumed." A compass you can defend gives Player 2 a reason to hand you data.

■ THE 12-YEAR-OLD WARNING

Profit equals revenue minus cost, revenue equals price times units, and cost splits into fixed and variable. The former McKinsey engagement manager said a 12-year-old could recite that on command, so it earns you nothing on its own. Add color: state what you suspect and what you would check before anything else. Suppose the prompt were falling profit. "My guess is that units are slipping at dealers while price holds near \$2,000, because the private-label bike took the entry-level buyer. I'd check units by channel and by line before touching cost." Player 2 now has a claim to push on.



■ THE WASATCH WHEELS GROWTH MAP

In the table, you have run the renumber on the fixed map and moved Line Economics up to second, because you need margin by line before you decide where growth money goes.

REGION	QUEST	SO THAT
1. Market	E-bike units per year in each of the 25 countries	we know where demand sits
	Unit growth by country over three years	we push the momentum markets
	Speed and power rules in the 24 other countries	we know which markets need a new spec
2. Line Economics	Contribution margin by line at the \$2,000 average price	we grow the line that earns
	Wasatch Wheels units by line over three years	we back the line with momentum
	Pricing power by line: units lost per 10% price move	we know which line can carry a price rise
3. Customer	Willingness to pay by segment against the \$1,500 to \$4,000 range	we know whether to raise price or add a cheaper model
	Age, income, and household type of buyers by line	we find the riders the brand misses
	Repeat and referral rate among current owners	we turn owners into sellers
4. Company and Channels	Capability gaps against the top three rivals in battery, software, and dealer support	we know what to build before we scale
	Acquisition cost and lifetime value of a buyer by channel	we spend where a rider costs least and stays longest
	Taiwan assembly capacity above 400K units, plus cash to add a line	the plan is fundable

■ DELIVERING IT IN THREE MINUTES

Talk from the page. The order:

- 1** Headline: the objective, your assumption if you made one, and the four regions. "No target yet, so I'll frame this as finding the next \$100M, one sixth of today's \$600M, across four regions: market, line economics, customer, and the company's channels and capacity."
- 2** Regions in priority order. Name each, give one sentence per quest with its number and its "so that," and move on. "First, the market. I'd count e-bike units per year by country, so that we know where demand sits."
- 3** Close with your compass: the region you would start in and why, and ask Player 2 to begin there.

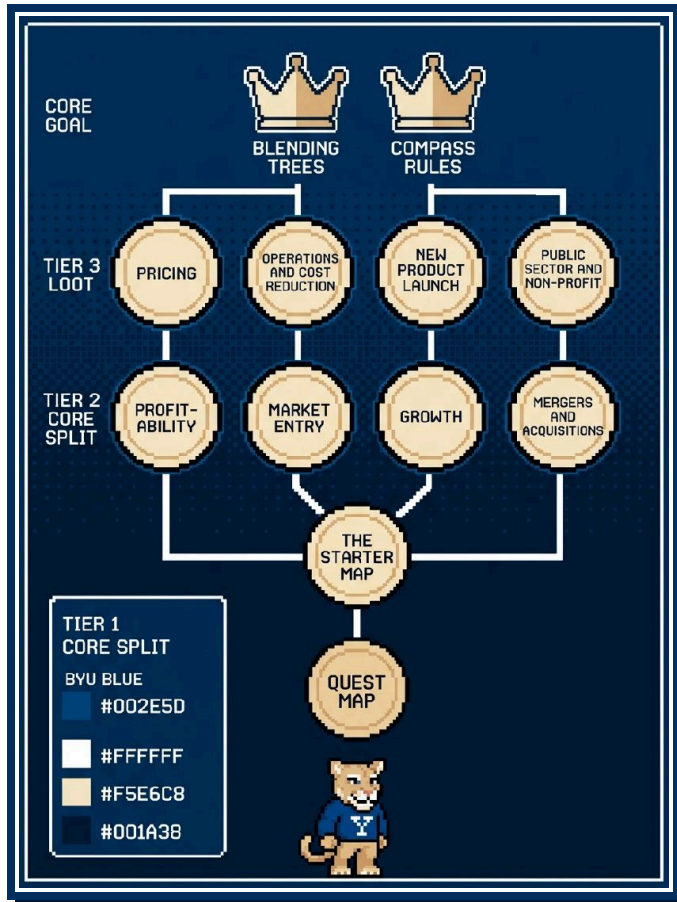
You run past three minutes when you have not ranked anything, because then you have to say all of it.

■ TIMING BENCHMARKS

Successful McKinsey candidates averaged about 2:30 head down and 2:15 to 3:00 on the walkthrough, and Bain interviewers like it snappier. Player 2 holds no stopwatch. Each minute you spend here comes out of the Boss Fights that follow, and the Boss Fights hold the Loot.

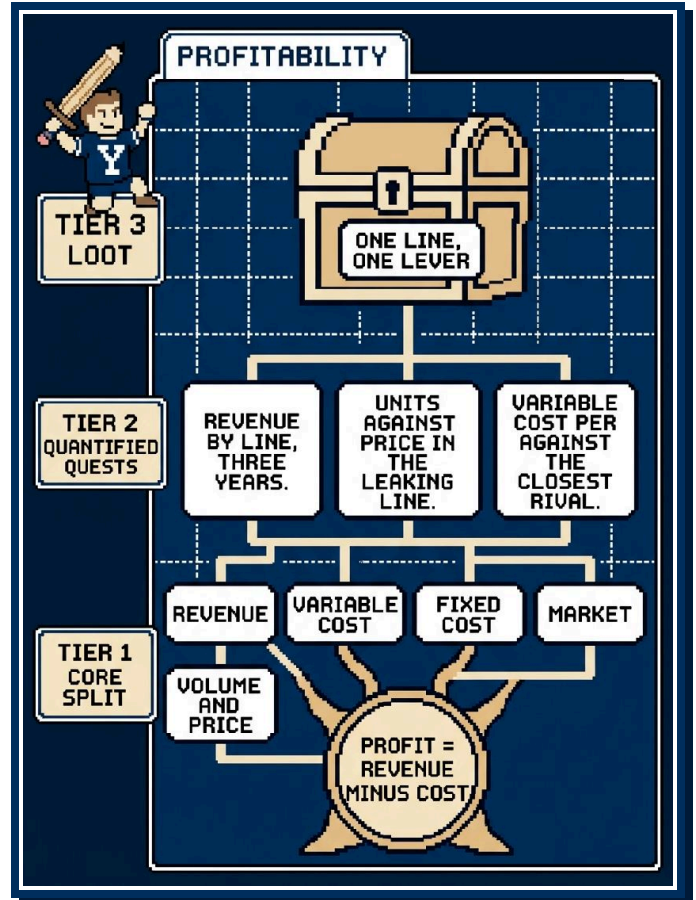
10 FRAMEWORK SKILL TREES

A skill tree is a Quest Map you built before the Boss Fight started. Tier 1 is the core split, the regions you draw in the Pause Menu without effort. Tier 2 holds the quantified quests, each tied to a "so that" the client cares about. Tier 3 is the Loot you expect to hand Dana Okafor. You unlock a tree by running it out loud on five or six cases until Tier 1 comes without thought and your two minutes go to Tier 2. Player 2 has heard the bare tree before, so the trees below wear Wasatch Wheels colors.



THE STARTER MAP

Draw Market, Product, Customer, Company when the prompt hands you no obvious tree. Market holds size, growth, and the competitor set. Product holds the lines, their features, and their economics. Customer holds the segments, what they want, and what they pay. Company holds cost, capacity, and channels. The coaches praised a student who opened an unfamiliar prompt with these four, because the regions cover most business problems and hold no overlap. Reach for it on a broad prompt with no clear root, and drop it the moment a named tree fits: a falling-profit prompt wants the profit tree, and a buy-or-pass prompt wants the M&A tree.

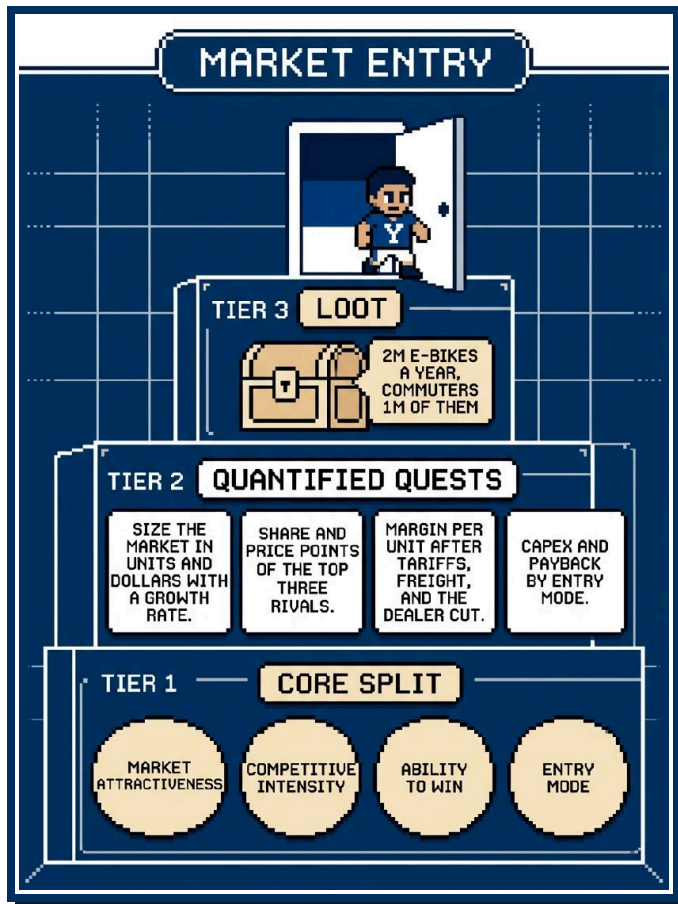


PROFITABILITY

Tier 1. Profit equals revenue minus cost, and you draw four regions around it. Region one splits revenue by line, channel, and geography, separating unit volume from price inside the line that leaks. Region two is variable cost per unit. Region three is fixed cost. The fourth region is the market, meaning growth and competitor moves, so that you can tell a company problem from an industry problem.

Tier 2. Pull revenue by line for three years, so that you find the line that is leaking. Split units from price inside that line to learn whether volume fell or price gave way. Compare variable cost per unit with last year and the closest rival, because a supplier or freight problem shows up there.

Tier 3. You name one line and one lever. In the Wasatch Wheels exhibit, recreation came in at -10% while cargo posted +50%. If recreation lost units at a held price, you suspect a rival's price cut or a demand shift, and each calls for a different fix.



■ MARKET ENTRY

Tier 1. You draw four regions: market attractiveness, competitive intensity, the client's ability to win, and the mode of entry with its economics.

Tier 2. Size the market in units and dollars with a growth rate, so that you know the prize. Measure the share and price points of the top three competitors to learn whether you fight one incumbent or a fragmented field. Compute contribution margin per unit after tariffs, freight, and the dealer's cut, because a sale there has to make money. Estimate capex and payback for building, partnering, or buying your way in, then rank the modes by those two numbers.

Tier 3. You give a binary answer with a line and a mode attached. In the German case you compute 2M e-bikes per year with commuters at 1M of them, so the commuter line leads the entry through dealer partnerships.

■ GROWTH AND THE PRODUCT-MARKET MATRIX

Tier 1. You cross current or new product with current or new market and get four cells: sell more of today's bikes to today's buyers, carry today's bikes into new geographies or segments, put new products in front of today's buyers, and the far corner where both are new.

Tier 2. Measure share of wallet at the 1,200 dealers, so that you know how much penetration remains at home. List the countries where the client sells but holds low share, then rank the near geographies by that gap. Count accessory and service revenue per bike to size the attach opportunity.

Tier 3. You rank the four cells by size of prize, time to cash, and risk, and let near cells fund far cells. Wasatch Wheels earns 70% of its revenue in the US and sells in 25 countries, so the second cell holds the largest untapped prize.

■ MERGERS AND ACQUISITIONS

Tier 1. You draw four regions: standalone value of the target, synergies with the buyer, price and deal structure, and integration risk.

Tier 2. Compare the target's growth and margin with the client's, so that you know whether you are buying growth or repairs. Map the overlap of dealers and customers to size revenue synergy without double counting. Estimate shared procurement and plant consolidation as a share of combined cost, which gives you the cost synergy. Set the price against standalone value plus synergies, because that sum is the most the client should pay.

Tier 3. You say buy or pass at this price and name a walk-away number. If Wasatch Wheels looks at a battery maker, the Loot is whether secured supply and lower cell cost cover the premium, and the least-supported number will be the cost synergy.

■ PRICING

Tier 1. Cost sets the floor, customer willingness to pay sets the ceiling, and competitor prices set the reference point. A fourth region covers architecture and channel: list price, discounts, financing, bundles, and dealer reaction.

Tier 2. Compute variable cost per unit by line, so that you know the floor and how much room today's price leaves above it. Measure willingness to pay by segment through a price-versus-features survey to find the ceiling. Estimate units lost per 10% price move, which gives you a working elasticity.

Tier 3. You give a price, a rationale, and a test. Cargo posted +50% and its buyers haul children, which points to low price sensitivity, so you test a modest increase there and watch units. A big-box private label sits 40% below the average price, so you hold on price support until the dealer sell-through data is in.

■ OPERATIONS AND COST REDUCTION

Tier 1. Lay the cost base out along the flow: procurement, assembly, logistics, and overhead. Inside each region you pull three levers: the price you pay, the quantity you use, and the efficiency of the process.

Tier 2. Compare cost per unit by component with a benchmark, so that you find the widest gap. Check supplier concentration and the savings a sourcing program could deliver, with 5% to 15% as the usual range, to size procurement. Measure OEE on the assembly lines against 60% typical and 85% world class, because the gap is the capacity you lose to downtime.

Tier 3. You hand over savings ranked by size, ease, and time, with quick wins apart from structural moves. The US plant question is this tree in miniature: a net annual benefit of \$30M, payback in 3 years, breakeven at 200K units, and tariffs as the assumption that could break it.

■ NEW PRODUCT LAUNCH

Tier 1. You draw four regions: customer need and market size, product fit including cannibalization, launch economics, and go-to-market.

Tier 2. Size the segment the product serves, so that you know the prize. Estimate the share of buyers who would switch from an existing line, then count net new units. Compute contribution margin per unit and fixed launch cost, because those two numbers give you breakeven units.

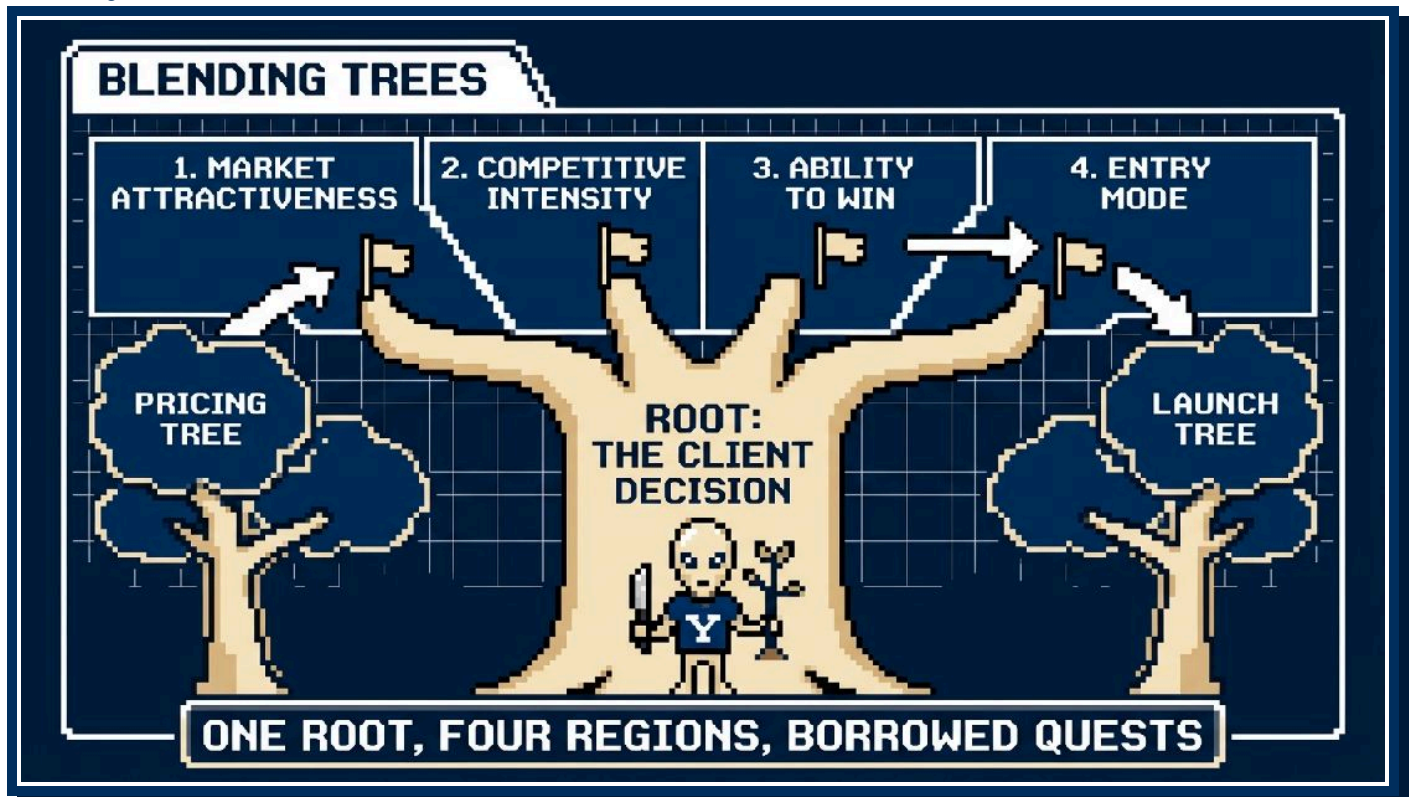
Tier 3. You say launch, delay, or kill, anchored on breakeven units. A folding commuter bike that needs 10K units to break even in a segment that buys 100K a year, one bike in ten, is a launch that opens with a dealer pilot. A bike that needs half the segment is a kill.

■ PUBLIC SECTOR AND NON-PROFIT

Tier 1. You swap profit for mission outcome. The regions become the outcome and its measure, the beneficiaries and stakeholders, funding and cost, and feasibility, which covers politics and staff capacity.

Tier 2. Track the outcome metric and its trend, such as officers hired each year against officers leaving, so that you know the gap. Compute cost per outcome by program to rank programs by impact per dollar. Break funding out by source and trend, because the line at risk shows up there.

Tier 3. You rank the options on impact per dollar and feasibility. For a philanthropy chasing younger donors, you size donors under 40 as a share of gifts and test the channel with the lowest cost per new donor.



■ BLENDING TREES

World 2 prompts often blend two frameworks. Dana Okafor might ask whether Wasatch Wheels should enter Germany with a new cargo model, and at what price. You draw one map and take its root from the client's decision. The decision is binary, enter or stay out, so market entry is the trunk and you borrow branches from the other trees. Product fit and cannibalization

from the launch tree become quests inside the ability-to-win region. Willingness to pay from the pricing tree becomes a quest inside market attractiveness, and competitor price points become a quest inside competitive intensity. The rule holds across hybrids: one root, four regions, and any second framework donates quests to those regions.

■ COMPASS RULES

State your Compass as soon as the map is on the table: "My hypothesis is that growth comes from the commuter line in the US and Germany, so I would start in region one." Player 2 can then steer you and can see you hunting for an answer.

Run 80/20. You will get most of the answer from two or three of the twelve quests. Pick the quest whose result would change the recommendation and ask for that data before the rest.

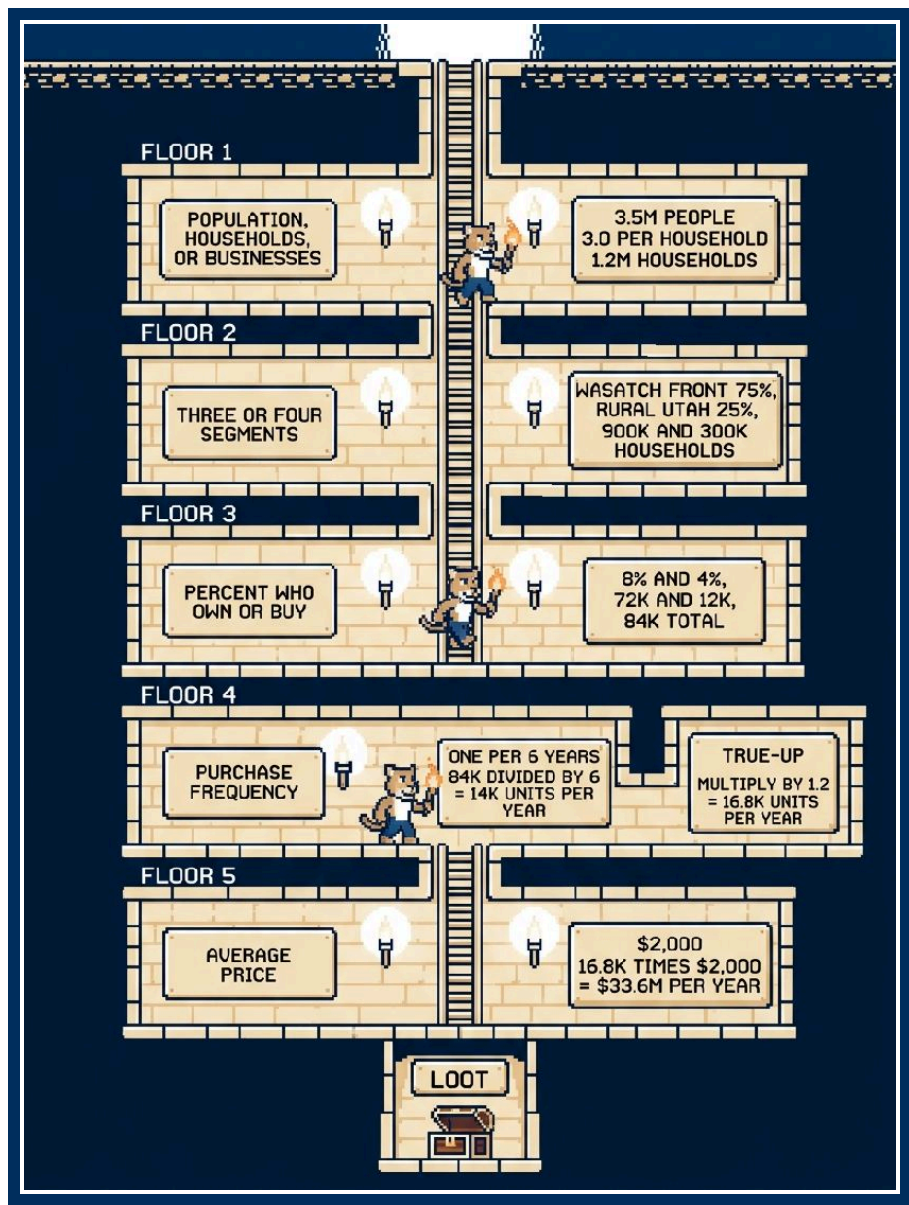
Do not boil the ocean. You will run two or three quests in a thirty-minute case, and Player 2 will hand you the exhibits that matter. Insist on touring all twelve and you ramble past three minutes and lose the room. If the first quest kills your hypothesis, say so out loud, turn the Compass, and start the next region. Player 2 scores you higher for a hypothesis you stated and dropped with a reason than for a case you ran without one.

11 LEVEL 3: THE SIZING DUNGEON

The Sizing Dungeon appears as a mini case in a screening call or as one room inside a full case. Player 2 names a product and a place, and you clear five floors in order, narrating each one.

■ GROUND RULES BEFORE YOU DESCEND

Market size means revenue in dollars per year unless Player 2 says otherwise. Profit needs a margin, and Player 2 did not hand you one. Confirm the unit at the Save Point, ask for the Pause Menu, sketch the five floors as a tree, and walk Player 2 through it before you touch a number. Player 2 grades each step you speak once the structure passes.



THE FIVE FLOORS

Floor 1: Population, households, or businesses. Pick the unit that opens its wallet: a cargo bike belongs to a household, a helmet to a head, a courier fleet to a business.

Floor 2: Three or four segments. Choose groups that buy at different rates or prices.

Floor 3: Percent who own or buy. Assign one share per segment with a one-line reason.

Floor 4: Purchase frequency. Count purchases per owner per year: below one for durable goods, above one for consumables.

Floor 5: Average price. Set one price per segment if the segments buy different products.

Multiply down the floors, add the segments, true up for the buyers your unit missed, and hand over the Loot: what the size means for the client.

SEGMENTATION IS AN ART

Six cuts cover most markets. Age works when the product changes with the decade of life: kids outgrow helmets, retirees stop commuting. Income works when price drives the decision. Choose geography (urban, suburban, rural) when terrain or density shapes demand. Gender works for personal-care categories and adds little for most vehicles. Needs or use case works when one product means different things to different buyers. Sunscreen is a beach product, a sport product, or daily skincare, and an e-bike is a commute tool, a trail toy, or a family hauler. Cut by culture or ethnicity when it drives demand and you speak about it with respect: a hair-care retailer built on textured-hair products, a quinceañera dress shop where those families live.

Any cut must pass one test: the segments differ on Floor 3, 4, or 5.

■ DURABLE GOODS AND THE REPLACEMENT CLOCK

A durable good lasts more than a year, so an owner buys less than once a year. An e-bike on a 6-year cycle means one owner in six buys this year: 1/6, or 16.7%, off the Fraction Scroll. Convert years to a fraction before you multiply.

■ THE STOCK VERSUS FLOW TRAP

The most common Game Over in this dungeon: you multiply owners by price and announce the market. That number values the units sitting in garages, a stock. Market size is a flow, the purchases made this year. One student at the workshop reached a stock number, caught it out loud, and the coach counted the catch in his favor. Press Continue by narrating the catch: "That gives me 84K e-bikes in homes, the installed base. I need annual purchases, so I divide by the replacement cycle."

PLAYER 2

I'm curious about the 8% on the Wasatch Front. Talk me through it.

COSMO

Two anchors. About 5M Americans own an e-bike against 130M households, so about 4% of households have one. The Wasatch Front has the trails and the commuter income, so I doubled that to 8%. Rural Utah stays at the national 4%. If you have a better figure, I'll swap it in.

Three answers like that unlock the Reasonable Numbers trophy.

■ THE TRUE-UP

A household model misses buyers who enter through a different door: courier fleets, resort rental shops, campus bike shares. Add 10% to 20% and name what the add covers, because the reason matters more than the digit. The Utah walkthrough uses 1.2 because resort towns rent e-bikes by the hour and couriers ride them all day.

■ THE REASONABLE TEST

Player 2 grades the logic and treats the number as a byproduct. Two students at the workshop sized one market with assumptions far apart, and both passed because the logic held. The coaches said the true figure could be ten times bigger or smaller and nobody in the room knows. One assumption fails on sight: 100% ownership.

The probe arrives as "why 60%?" and the answer has three parts: an anchor you can observe, an adjustment you can defend, and an offer to swap in a better figure.

■ WALKTHROUGH: E-BIKES SOLD IN UTAH PER YEAR

You segment Utah by geography, because commute patterns split the state.

FLOOR	STRUCTURE	ASSUMPTION	CALCULATION
1	Utah households	3.5M people, 3.0 per household	1.2M households
2	Segment by geography	Wasatch Front 75%, rural Utah 25%	900K and 300K households
3	Own an e-bike	8% on the Wasatch Front, 4% in rural Utah	72K and 12K, 84K total
4	Replacement purchase	one per 6 years	84K divided by 6 = 14K units per year
True-up	Fleets, rentals, businesses	multiply by 1.2	16.8K units per year
5	Average price	\$2,000	16.8K times \$2,000 = \$33.6M per year

State the Loot: \$33.6M a year against \$600M of Wasatch Wheels revenue means owning all Utah sales covers under 6% of the top line. Growth lives outside the home state, so the German question in Boss Fight I carries weight. Your next check is Wasatch Wheels' current share of Utah sales.

■ SECOND RUN: SKI HELMETS SOLD IN COLORADO

A helmet belongs to a person, and age drives replacement and price, so you segment Colorado's 6M people by age. The 1/80 trick puts people under 18 at 18 times 1.25%, about 22.5%. You round to 25% for clean math and say so. You take 65 and over at 15% (0.9M) rather than the trick's 19%, because Colorado skews young, which leaves 60% (3.6M) for 18 to 64.

First the funnel, from people down to helmets owned.

SEGMENT	POPULATION	SKIERS	OWN A HELMET
Under 18	1.5M	20%, 300K	90%, 270K
18 to 64	3.6M	25%, 900K	80%, 720K
65 and over	0.9M	10%, 90K	80%, 72K

Then the money, from helmets owned down to revenue.

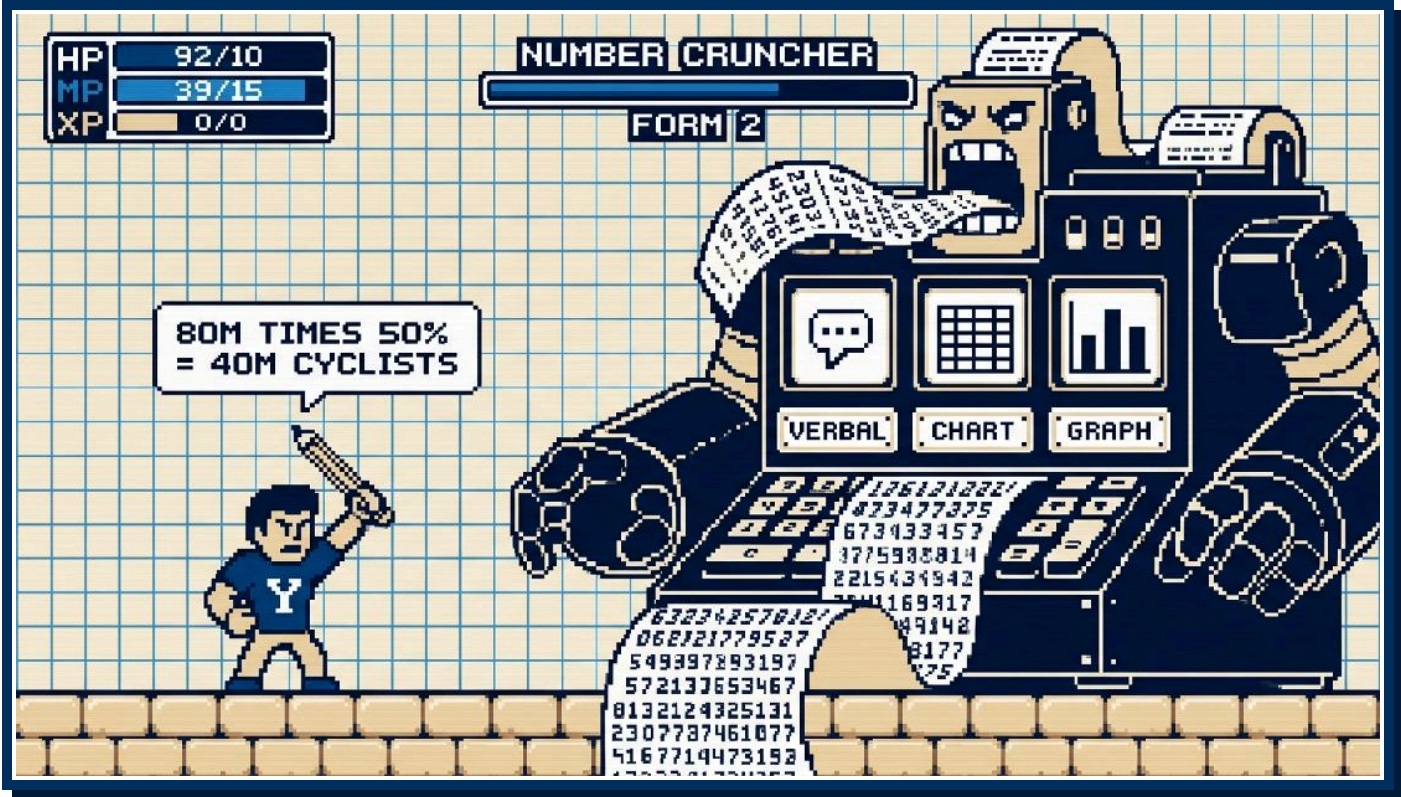
SEGMENT	CYCLE	UNITS A YEAR	PRICE	REVENUE
Under 18	3 years	90K	\$60	\$5.4M
18 to 64	6 years	120K	\$120	\$14.4M
65 and over	6 years	12K	\$100	\$1.2M

Residents buy 222K helmets for \$21M a year. A 20% true-up for out-of-state visitors and rental fleets brings it to \$25.2M. Player 2 will probe the 25% adult participation, and you have the anchor ready: half the state lives in the Denver metro (3M), within two hours of a dozen resorts.

THE MAP BECOMES THE MODEL

In the team room, the tree you drew in two minutes becomes the spreadsheet: each floor a column, each segment a row, each guess a highlighted cell. Researchers swap the highlighted cells for census counts, surveys, and dealer data, and the structure stays. Your day-one job is to sell three teammates on the shape of the model. A reasonable number clears the dungeon, and a structure the team can build on earns the offer.

12 BOSS FIGHT I: THE NUMBER CRUNCHER



The Number Cruncher spawns after the Quest Map and respawns two or three times before the close. This is the one phase of the Boss Fight with a single right answer, and it does its damage through pressure: the arithmetic is easy and the audience is not.

THE THREE FORMS

The boss takes three forms. In verbal form, Player 2 reads a story problem and your notes are the only record. In chart form, you get a table such as a P&L. In graph form, you get a bar chart or a line. The form changes the input. Your process stays the same.

THE FOUR STEPS

STEP	YOU	TIME
1. Recap	State what you are solving for, then read the data back by segment with units confirmed.	30 seconds
2. Structure	Write the equation in words before you touch a number. Ask for anything missing. Get the nod.	About 2 minutes
3. Solve	Work one step at a time, out loud, and check each result against the last.	About 2 minutes
4. Insight	Give the three-part answer: the number, the Loot, the next step.	About 1 minute

Structure first and get the nod. Say the equation in words, then stop and ask Player 2 whether the approach works. If you skip a step, Player 2 says so before you spend three minutes on the wrong arithmetic. At many firms the nod ends the question: a former McKinsey engagement manager told the room that full arithmetic takes three to four times as long as the setup, so an interviewer may take your structure and move on. Build the structure as if it were the whole answer.

Solve out loud. Narrate each multiplication and division as you write it. You are in co-op with Player 2, and four minutes of silence leaves your partner nothing to grade. After each step, compare the result with the one before it. A segment larger than its parent population means a decimal slipped, and Player 2 credits the catch.

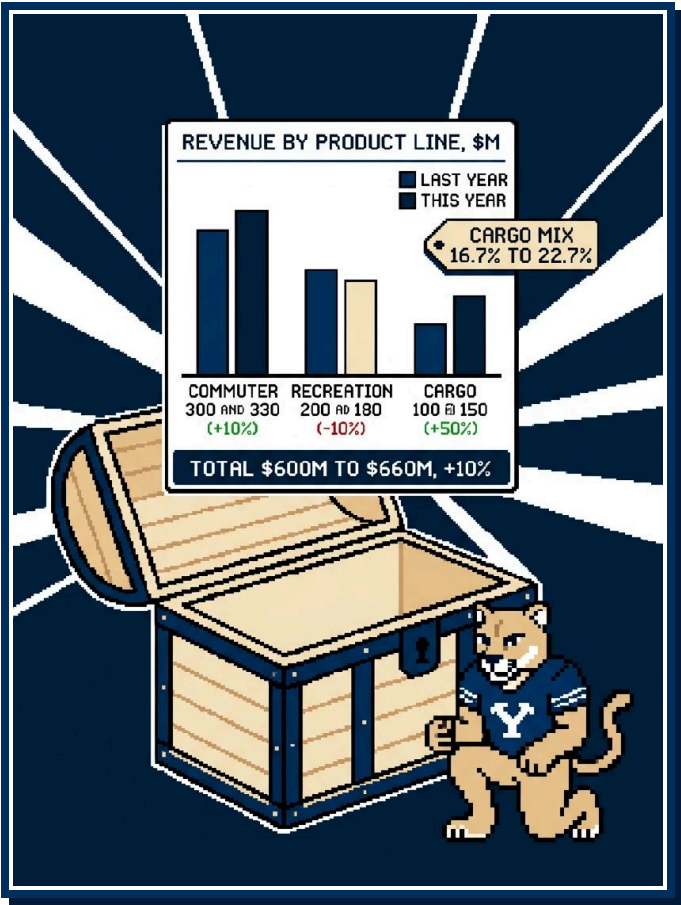
The three-part answer. A number without Loot is an empty chest. Part one is the number, in the units Player 2 asked for. Part two is the Loot: what the number means and what the client should do about it. Part three is what comes next: the next analysis and the risk that could flip the answer. Executives pay you to hide the mountain of data and hand over what matters.

■ WALKTHROUGH: THE GERMAN EXPANSION

Player 2 reads the verbal form: Germany has 80M people (84M real), and half of them buy a bicycle in a given decade. Buyers split casual 60% on a 12-year cycle, commuter 30% on 6 years, and enthusiast 10% on 4 years. E-bikes are 25%, 50%, and 50% of those purchases. Player 2 wants bikes per year, e-bikes among them, and the meaning for the client.

STEP	CALCULATION	RESULT
Cyclists	80M times 50%	40M cyclists
Segments	60%, 30%, and 10% of 40M	Casual 24M, commuter 12M, enthusiast 4M
Bikes per year	24M divided by 12, 12M divided by 6, 4M divided by 4	2M, 2M, and 1M, for 5M bikes total
E-bikes per year	25% of 2M, 50% of 2M, 50% of 1M	0.5M, 1M, and 0.5M, for 2M e-bikes total

Sense check the total before you move on. 5M bikes across 40M cyclists means one new bike per rider in eight years, which sits between the casual and commuter cycles, so the total passes. The Loot: commuters are 1M of the 2M e-bikes, half the market by units from 30% of the riders, so the commuter line leads the entry. Then the follow-ups: German competitor share and price points, the dealer network, EU speed and power regulations, and whether Germany beats the other candidate countries.



■ READING EXHIBITS

In chart and graph form the numbers sit in front of you, and the boss tests whether you read them in order. Run this checklist before you speak:

- 1 Title: the exhibit's claim.
- 2 Axes: what runs across and what runs up.
- 3 Units: dollars or units, millions or thousands, percent or points.
- 4 Time period: one year, five years, one quarter.
- 5 The biggest bar: the largest value and its share of the total.
- 6 The trend: up, down, or flat, and by how much.
- 7 The outlier: the row that moves against the others.
- 8 The so what: one sentence Player 2 could repeat to the client.

Player 2 slides one exhibit across, titled REVENUE BY PRODUCT LINE, \$M.

PRODUCT LINE	LAST YEAR	THIS YEAR
Commuter	300	330
Recreation	200	180
Cargo	100	150

Start with the column totals: \$600M to \$660M, +10% overall. Then growth by line: Commuter +10%, Recreation -10%, Cargo +50%. Then mix: Cargo mix rose from 16.7% to 22.7%, and you read the first figure off the Fraction Scroll, since 100 of 600 is 1/6. The outlier is recreation, the only line shrinking. The Loot: growth is coming from cargo while recreation shrinks, and the question to ask next is margin by line before you pour money into cargo.

■ WALKTHROUGH: THE ASSEMBLY PLANT

Player 2 reads: Wasatch Wheels imports frames and assembles in Taiwan. A US plant would cost \$90M to build, add \$30M per year in fixed operating cost, and save \$150 per unit in tariffs and freight. Volume is 400K units per year. Should the client build it?

Structure in words: saving per unit times volume, minus the new fixed cost, gives the net annual benefit. Capex divided by that benefit gives payback. Fixed cost divided by saving per unit

gives breakeven volume. Then:

- Annual saving: 400K times \$150 = \$60M
- Net annual benefit: \$60M minus \$30M fixed = \$30M
- Simple payback: \$90M divided by \$30M = 3 years
- Breakeven volume: \$30M divided by \$150 = 200K units per year

Simple payback ignores the time value of money, so a dollar saved in year three counts the same as a dollar saved this year. It also ignores what the plant saves after year three, where the rest of a long-lived asset's value sits. If Player 2 pushes, run net present value at the client's hurdle rate across the plant's life and an internal rate of return to set beside that hurdle.

The Loot: the plant pays back in 3 years at current volume and still breaks even if volume halves. The risk is the assumption that tariffs stay where they are, since the \$150 saving carries the case: at \$100 per unit, the net benefit drops to \$10M and payback stretches to 9 years. Next steps: a tariff scenario from trade counsel and capex quotes from two builders. The question is binary, so the answer is binary: build, and name the tariff risk in the same breath.

■ HOCK DIALOGUE: THE GERMAN BOSS

PLAYER 2

Let's test your first region with numbers. Wasatch Wheels wants to enter Germany. Use 80M people. The real figure is 84M. Half the population does not buy a bicycle in a given decade. Of the people who do buy, 60% are casual riders on a 12-year replacement cycle, 30% are commuters on a 6-year cycle, and 10% are enthusiasts on a 4-year cycle. E-bikes are 25% of casual purchases, 50% of commuter purchases, and 50% of enthusiast purchases. I want bikes bought per year, e-bikes among them, and the meaning for the client.

[COSMO WRITES AS PLAYER 2 TALKS]

COSMO

Let me read that back by segment. 80M people, half of them buy. Casual: 60%, one bike per 12 years, 25% electric. Commuter: 30%, one per 6 years, 50% electric. Enthusiast: 10%, one per 4 years, 50% electric. You want annual units, bikes then e-bikes.

PLAYER 2

Correct.

COSMO

One check: I read the e-bike shares as a slice of each segment's bikes, inside the bike total.

PLAYER 2

Inside the total.

COSMO

I'd like a minute to set up the math.

**PLAYER 2**

Take it.



[COSMO WRITES FOR 60 SECONDS]

COSMO

Approach first, then numbers. Population times the buying share gives cyclists, split three ways by segment. Each segment divided by its replacement cycle gives bikes per year, summed. Each segment's bikes times its e-bike share gives e-bikes, summed. Does that setup work for you?

**PLAYER 2**

It does. Run it.

**COSMO**

80M times 50% is 40M cyclists. Segments: 60% of 40M is 24M casual, 30% is 12M commuter, 10% is 4M enthusiast. Those sum to 40M, so the split holds.

**PLAYER 2**

Good.

**COSMO**

Next I'll take bikes per year. Casual: 24M over 12 is 2M. Commuter: 12M over 6 is 3M. Enthusiast: 4M over 4 is 1M. That gives 6M bikes a year.

**PLAYER 2**

Go on.

**COSMO**

Before I do, one check. Commuters have half the riders casual has and replace twice as fast, so the two segments should land on the same number. 12 divided by 6 is 2, so the commuter line is 2M a year and the total is 5M bikes.

**PLAYER 2**

Agreed. Keep going.

**COSMO**

Then the e-bike step. Casual: 25% of 2M is 0.5M. Commuter: 50% of 2M is 1M. Enthusiast: 50% of 1M is 0.5M. Total 2M e-bikes a year.

**PLAYER 2**

So the number is 2M. Tell me what Dana does with it.



COSMO

I have three things for her. First, the size: Germany buys 5M bikes a year and 2M of them are e-bikes, about five times Wasatch Wheels' current global volume, so the market can move our numbers.

**PLAYER 2**

Five times current volume. Second?

**COSMO**

Second, the Loot: commuters buy 1M of the 2M e-bikes, half the market from 30% of the riders. The commuter line should lead the entry.

**PLAYER 2**

And the third?

**COSMO**

Third, next steps: German competitor share and price points, the dealer network, EU speed and power rules, and whether Germany beats the other candidates.

**PLAYER 2**

Casual riders are 24M people, twice the commuter pool. Sell me on leaving the biggest crowd for later.

**COSMO**

I would sequence casual second, and the purchase clock is the reason. A casual rider buys an e-bike about once in 48 years and a commuter about once in 12, so the commuter clock runs four times faster. Commuters ride on weekdays, where range and reliability sell. Casual is wave two, once dealers carry us.

**PLAYER 2**

You said the market can move our numbers. That assumes German commuters pay what US riders pay.

**COSMO**

Agreed, and price changes the revenue case without changing which segment buys the most units. I would put a German price scan at the top of the next steps, ahead of any revenue estimate.

**PLAYER 2**

Good. Let's look at an exhibit.



13 BOSS FIGHT II: THE IDEA HYDRA



The second boss can arrive before or after the math and carries no numbers. Player 2 asks an open question on a narrow topic. Cut one head and two more appear. You win by naming three necks first and growing heads in an order you control.

■ A MINI QUEST MAP

The Idea Hydra is the Quest Map shrunk to a four-minute window. You split a pointed prompt (a retailer undercut us, tell me what to look into) into three necks with two to four heads each after 30 to 60 seconds. Player 2 grades the same three things in both fights: categories that cover the question without overlapping, ideas that fit this client, and the Loot, the action you name at the end. The shorter prep means thinner cupboards, and the coaches said Player 2 expects that.

Player 2 holds no answer key for this fight. A different set of defensible necks scores the same, and the coaches said it twice about creative answers: math has one right answer, and the rest of the case is an art graded on coverage.

■ THE FOUR STEPS

The four steps take three to four minutes in all:

- 1 Recap and confirm, 30 seconds. Restate the question in one sentence, confirm the scope, and ask for 30 to 60 seconds.

- 2 Name the necks, 1 minute. Say the three categories out loud before you fill any of them.
- 3 Grow the heads, 1 to 2 minutes. Two to four ideas per neck, each specific to this client.
- 4 Start and what else, 30 seconds. Pick the head you would chase first and say why. Name one thing you left out.

Two rules run this fight. Expect "And what else?" and answer it by going one level deeper on a neck you already named or by opening a fourth neck. Never restate a used head. Treat the push like a dodge, a request for depth. And turn topics into checks: "battery quality" is a topic, while "tear down the battery and compare cell supplier against ours" is a head Player 2 can picture a team doing.

■ THE PRIVATE-LABEL PROMPT

The prompt: a national big-box retailer has launched a private-label e-bike at 40% below the Wasatch Wheels average, about \$1,200 against your \$2,000, and dealers want price support. Player 2 asks what you would investigate before the client responds.

NECK	HEADS
Product and quality	Specs against the commuter line, battery cell supplier and cycle rating, warranty and who services a claim, who builds the bike
Economics	Whether the retailer sells below cost to pull traffic, how long it can hold that price, the supplier's terms and volume commitment
Customer and channel	The segment that is switching, the weight of brand in this category, the dealer view across all 1,200 dealers

Your starting point: mystery-shop the bike this week and pull dealer sell-through for the last 90 days. Your what-else: a dealer-exclusive model and a financing offer, two responses to test before any list-price cut.

■ HOCK DIALOGUE

PLAYER 2

One more before we wrap. The private-label bike is at \$1,200, about 40% under our average, and dealers are asking Dana for price support. Before she answers them, tell me what you would investigate about the retailer's move.

COSMO

So we need to learn about the retailer's bike and its intent before Dana picks a response, and price support is one option among several. Correct?

PLAYER 2

Correct.

COSMO

Can I take 45 seconds to organize?

PLAYER 2

Go ahead.

[COSMO WRITES THREE HEADERS AND FILLS UNDER THEM FOR 45 SECONDS]

COSMO

Three areas: product, economics, and customer and channel. I will take them in that order and then say where I would start.

PLAYER 2

Go.

COSMO

Product. Specs side by side with our commuter model. A battery teardown, because the battery is the largest cost in an e-bike. The warranty and whoever services a claim, since few big-box stores keep a bike mechanic. And who builds it, because one Taiwanese contract assembler could be building both bikes.

PLAYER 2

Fine. Economics.

COSMO

Whether the retailer makes money at \$1,200 or uses the bike as a traffic driver the way grocers use milk. If it is a loss leader, how long they can carry it depends on the supplier's terms and any volume commitment.

**PLAYER 2**

Keep going.

**COSMO**

Customer and channel. The segment that is buying: first-time riders who would not have paid \$2,000, or our commuter customers trading down. The weight of brand in e-bikes, which I suspect is heavier than in a toaster because of service and resale. And the dealer view: sell-through by model over the last 90 days.

**PLAYER 2**

Good list. And what else?

**COSMO**

Under product, safety certification on the battery and charger, because a recall on their side removes the price threat and the dealers' request with it. Under economics, the retailer's margin on helmets and service plans, since a bike sold at cost can still earn on accessories. Under channel, whether the bike sits on the retailer's website or on the floor of all its stores, since the second overlaps more of our dealer map.

**PLAYER 2**

Useful. And what else?

**COSMO**

One area I have not touched: timing, scale, and scope. A March launch looks like a spring promotion, and a year-round assortment looks like a permanent category. The order quantity, because 5,000 units is a test and 50,000 is a commitment. And whether this is one commuter model or the first of a cargo and trail range.

**PLAYER 2**

So where do you start?

**COSMO**

Two moves this week. Mystery-shop the bike and buy one for a teardown, then pull sell-through from our top 200 dealers for the last 90 days. If commuter sell-through held, the dealers are nervous and price support can wait. If it dropped, I know which segment moved.

**PLAYER 2**

Dealers are calling today. They want an answer.

**COSMO**

Then this week's answer is that we are testing two responses that protect them without cutting list price: a dealer-exclusive commuter model the retailer cannot match, and a financing offer that closes most of the monthly gap to the \$1,200 bike.



PLAYER 2

Good. Let's close.

**■ SECOND PROMPT: CARGO BIKES AT DEALERS**

Player 2 says: the cargo line is up +50% this year. On the website one order in four is a cargo bike. At dealers it is one sale in twelve. Dana wants ideas to raise the cargo share of dealer sales.

- Dealer incentives: cargo margin at or above commuter margin, a bonus per cargo sale, a demo-bike program, and a floor-space allowance.
- Customer experience: a test-ride loop with a weighted child seat, a staff script for parents who came in for a commuter.

- Offer: an entry cargo model near \$2,500, accessory bundles, family financing, and a trade-up credit for commuter owners whose families grew.

Your starting point: compare dealer margin and floor space per unit on a \$3,500 cargo bike against a \$2,000 commuter, then rank the 1,200 dealers by cargo share, visit the top 50, and copy what they do. Your what-else: map website cargo buyers against dealer locations, because if those buyers live nowhere near a dealer, the gap is geography and the fix is coverage.

14 FINAL BOSS: THE CHAIRLIFT



The last boss announces itself with a corny line. After 30 to 45 minutes of Quest Maps, exhibits, and hydra heads, Player 2 leans back and says one of the standard versions: "I have dinner with the CEO tonight and I need a concise message for her," or the BYU edition, "Dana Okafor is riding the chairlift with me this afternoon, one ride, about a minute." You lose HP here in two ways: you summarize the last question alone, or you freeze because the case has no tidy answer.

■ ASK FOR 30 SECONDS

Ask for the Pause Menu here the same way you asked for it before the Quest Map: "I have a page of numbers. Give me 30 seconds to put together a clean message for her." Trained interviewers expect the request and accept 30 to 60 seconds. Write three things on a fresh corner of the page: the recommendation in one line, the two or three numbers that support it, and the assumption you trust least. Do not reread your notes from the top. The chairlift leaves with or without you.

■ THREE STEPS ON ONE RIDE

The close has three parts, in this order.

- 1 Recap.** Say why the client hired you and what you did: one sentence on the problem, one or two on the work. Keep this under 20 seconds, because the CEO lived the problem and wants the answer.

- 2 Recommend.** Say what the client should do based on what you know now. A hedge with no direction is a Game Over screen, because firms hire people who commit on partial data and clients pay for a decision before the data is complete.
- 3 Next steps.** Name three areas to go deeper or wider: the analysis that would firm up your weakest number, the risk you would watch, and the question the case did not reach.

■ BINARY QUESTIONS GET BINARY ANSWERS

Some cases end on a yes-or-no question: buy the company or pass, enter Germany or stay home. Answer yes or no in your first sentence, then give the reasons and the next steps. Player 2 asked a binary question and grades the first word. For an open question, such as how the client should respond to a low-price rival, skip the yes or no and lead with the recommendation itself.

■ NAME THE WEAKEST NUMBER

One assumption in your recommendation carries more weight than its evidence. Name it before Player 2 does. In the German math, you assumed e-bikes are 50% of commuter purchases, counted 1M commuter e-bikes on that basis, and made the commuter line the lead. If that share is closer to 30%, the

commuter figure drops to 0.6M, the segment shrinks by 40%, and the German market you sized drops from 2M to 1.6M e-bikes a year. Then state the fix: a competitor pricing scan and a round of German dealer interviews before you commit the launch budget. Two more pairs from the workshop: a consumer survey on price sensitivity before you raise prices, and vendor quotes before you promise supply-chain savings. One coach called this the professional way to admit a number feels shaky.

■ DRAW ON THE WHOLE FIGHT

Build the recommendation from all of it. Pull one number from each phase: cargo grew +50% while recreation posted -10%, commuters buy 1M of the 2M e-bikes sold in Germany each

PLAYER 2

Dana Okafor is riding the chairlift with me this afternoon. One ride, about a minute. She wants to know what we recommend on growth, and I need to know what to tell her.

COSMO

I have a page of numbers. Give me 30 seconds to pull the message together.

PLAYER 2

Go ahead.

[COSMO WRITES FOR 30 SECONDS]

COSMO

Dana hired us to find growth for a \$600M company that sells 400K e-bikes a year through 1,200 dealers, and we sized Germany, read the product-line results, tested a US plant, and looked at the big-box private label. Three updates for her. First, the recommendation: lead the growth plan with the commuter line and enter Germany. Commuters buy 1M of the 2M e-bikes sold in Germany each year, and commuter revenue grew +10% at home. The US plant pays back in 3 years at current volume, so it can wait behind the German entry as a margin project. Hold on price cuts against the private label until the dealer sell-through data comes in.

PLAYER 2

Keep going.

COSMO

Second, the number I trust least is the 50% e-bike share of German commuter purchases. If it is closer to 30%, the commuter segment is 40% smaller and Germany drops from 2M to 1.6M e-bikes a year. Commuters keep the lead by a smaller margin.

PLAYER 2

Give me the next steps.

COSMO

Third, next steps: run a competitor pricing scan in Germany, pull dealer sell-through for the last 90 days, and get margin by product line before we commit money to cargo.

year, and the plant pays back in 3 years. Put those three side by side and Dana hears one story instead of three case questions.

■ THREE UPDATES

Speak the way a partner briefs an executive: "Three updates for you. Number one..." Lead with the answer, number the points, and stop. The Answer First trophy unlocks on this ride. If Dana wants detail on point one, she will ask.

■ MOCK FIGHT: THE WASATCH WHEELS CLOSE

PLAYER 2

Cargo grew +50% this year and you led with commuter. Defend that.

**COSMO**

Cargo is the fastest-growing line at home and 22.7% of revenue, with no margin data yet. Commuter is the largest line at home and half of the German market by units. Cargo moves to first if the margin work supports it.

**PLAYER 2**

Dana wants to cut prices tomorrow to match the big-box bike. Tell me why she should wait.

**COSMO**

Matching a 40% cut means 40% less revenue on each bike we sell at flat volume, about \$240M off a \$600M top line, spent before we know whether the private label has taken a single commuter sale. We get that answer from the sell-through pull in a week.

**PLAYER 2**

And if dealers are losing sales?

**COSMO**

Then we target the response. A dealer-exclusive model at a lower price point, or a financing offer through the 1,200 dealers, protects the segment under attack without repricing all three lines. I would test both before touching list price.

**PLAYER 2**

Good. That is the ride.



■ CHAIRLIFT GAME OVER SCREENS

The first screen is a two-minute recap that leaves no time for the recommendation. The second repeats the last question and forgets the first three. The third is a close with no next steps. Continue? Ask for 30 seconds, write three lines, and lead with the answer.

■ THE BLANK SCREEN

Mid-case, your mind empties. Say out loud what you are doing: "Let me take stock for a second." Reread your own objective line at the top of the page, name the last thing you established, and ask for ten seconds. Player 2 is co-op and would rather hear the stall than watch the silence, and the sentence you speak while stalling points you at the next quest more often than not.

15 SIDE QUESTS

Fit stories, networking, the other boss forms, and the AI trap decide whether you reach the chair and whether the team wants you once you do. Skip them and you can win the fight without

reaching the Victory Screen.



■ CHARACTER SELECT

Firms label the fit interview behavioral, fit, or a branded name such as McKinsey's personal experience interview. You will hold no formal authority inside the client: you cannot spend a dollar of the client's money or hire or fire anyone in the client's building. Your job is influence, and Player 2 wants proof that you have moved people who did not report to you.

McKinsey scores its personal experience interview on three themes: personal impact, entrepreneurial drive, and inclusive leadership. Tag each prepared story to at least one theme, and know which story you would lead with when Player 2 names the theme first.

Pull your backstory from missions, MBA projects, and jobs. One coach said many of the best leadership stories he heard as an interviewer came from missions. A companion who refused a new approach or a district whose numbers were sliding is a story about influence without authority. Class projects supply conflict and deadlines. Jobs supply dollar figures.

Build each story in five beats:

- Setup: one sentence on where you were and what your role was.

- Stakes: the problem, why it was hard, and one number if you have one.
- Moves: the decisions and actions you took, in first person singular.
- Result: one sentence with a number in it.
- Lesson: the move you would repeat or the move you would change.

Tell it in about two minutes. At some firms Player 2 then spends five to ten minutes inside the Moves beat asking why you chose that path. "We" hides you, so say "I." Prepare five or six stories, tag each to more than one prompt, and rehearse them out loud with a partner.

Three answers open or close most fit rounds, and you write them once and reuse them. The resume walk-through runs two minutes: the thread from your first job to this chair, one line per stop, ending on the reason the next stop is consulting. The reason for consulting names the work you want, the pace, the spread of problems, and the skill you plan to build. The reason for this firm runs on the specifics you collected in the

Overworld: the practice a second-year described, the office culture a partner named at the Tanner Building, the project a coffee chat put in front of you.

Player 2 then turns the round over and asks what you want to know. Bring two questions the firm's website cannot answer, and tie one of them to something Player 2 said. A question the recruiting page answers costs you the last two minutes of the round, and Player 2 files it as low interest.

Give fit the prep hours the case gets. Rubric question three is the Omaha test, so follow up on the small talk Player 2 opened with, and bring stories that prove you belong in that room.

■ THE OVERWORLD

Networking is travel across the map: coffee chats, campus events, NPCs who hand you the key to the chair. Once you sit down, the firm walls off the Overworld from the assessment, so a warm referral adds nothing to your case score.

Referral value swings by firm. Some firms take a partner's recommendation and put you in the chair. Others prohibit direct partner recommendations for campus candidates. At some, a note from a BYU MBA two years into the job carries weight, and one coach said that note is what gets you the interview. Ask each contact how referrals work at their firm.

MBA programs are the best-organized candidate pool a firm has, so partners come to the Tanner Building to shake hands and answer questions. One coach put in the miles as a candidate, collected zero formal referrals, and used those connections to land interviews. In each conversation, learn one thing the website does not list and ask who else you should talk to.

■ OTHER BOSS FORMS

Two things change what you do in the chair: the hand on the wheel, and the costume the firm puts on the boss.

The hand on the wheel. McKinsey runs interviewer-led cases: Player 2 picks the next question, hands you one piece of the case at a time, and expects a clean answer to the question on the table. Bain and BCG lean candidate-led: you drive from your own map, name the region you want next, and ask for the data that region needs. In an interviewer-led round, answer Player 2's question, stop, and wait for the handoff. In a candidate-led round, close each answer by naming where you go next and what you want to see.

The recruiter screen (World 0). A recruiter cut is rare if you speak in complete thoughts and laugh at their jokes. If the caller works in the business, expect a simple mini case such as sizing the e-bikes sold in Utah. Mini cases run 15 to 20 minutes, so walk the five floors of the Sizing Dungeon and land the number with its Loot.

The digital gate (World 0). Once you clear the bar, the firm sets the score aside. Some gates are reasoning games, some are a chatbot case, and some test numbers and charts under a clock, so read the firm's own prep page, then drill mental math and chart reading, the work that raises your MP.

The presentation interview. In the EY and PwC form, you sit alone with the deck, the spreadsheet, and no phone. Read the question first, write an answer-first storyline before you open the spreadsheet, and give each slide one message in its title. Panel questions are a Boss Fight in a suit: number, Loot, next step.

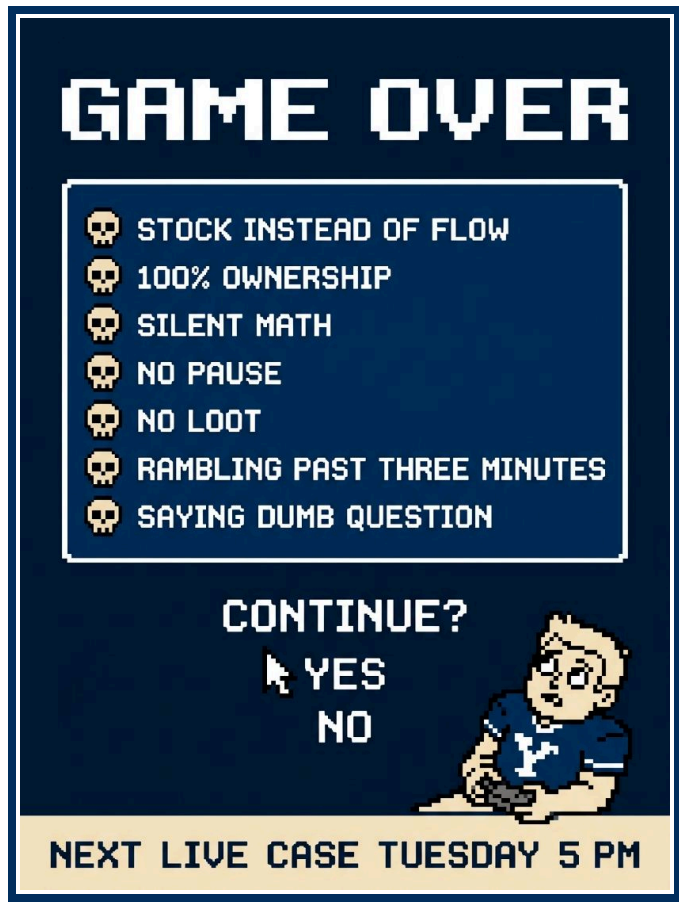
The group interview. Few MBAs see this form. Open with a structure, hand airtime to the quiet person, and tie the group's ideas together at the close.

■ THE AI TRAP

You can drill with AI and get sharper, or you can grade with AI and walk into the Dungeon blind. Good uses include mental math drills, Fraction Scroll quizzes, fresh prompts at 11 pm, building a Quest Map and asking the model to find the missing Region, and plain-language explanations of a sector new to you.

The trap is the feedback. One coach put it this way: the grading skews positive, and the model praises whatever you produce. A tool that calls your six-minute framework excellent has stolen the one thing practice exists for: finding the weak event in your decathlon.

Case with humans for the verdict. The best feedback comes from an MBA2 who went through consulting recruiting and has sat on both sides of the table for 20 to 30 cases. That person will tell you how you come across. Use AI for the drill and a person for the score.

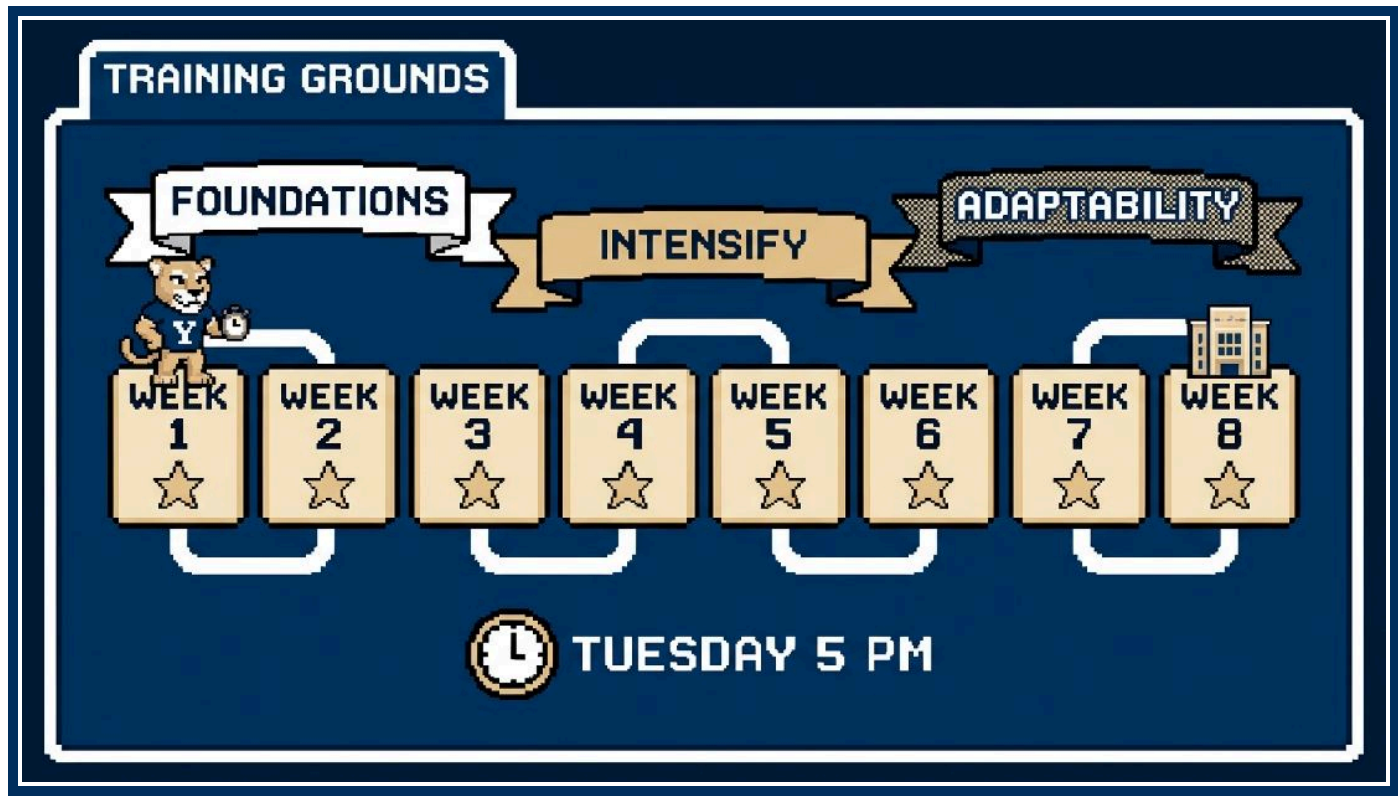


Game Over screens on the way:

- Stock instead of flow: you reported the 84K e-bikes in Utah homes when Player 2 asked for units sold per year.
- 100% ownership: you assumed a whole segment buys and failed the reasonable test.
- Silent math: you solved with your head down and Player 2 lost the thread.
- No pause: you built the framework without asking for the Pause Menu.
- No Loot: you delivered a number and stopped talking.
- Rambling past three minutes: your Quest Map ran long and lost its priority order.
- Saying "dumb question": Player 2 dodged, and you apologized for a question that belonged in your map.

Each screen ends the same way: "Continue?" Press yes. The next live case runs Tuesday at 5 pm.

16 TRAINING GROUNDS



The Boss Fight is a decathlon. A former McKinsey engagement manager told the room that a decathlete two weeks out, strong in the sprints and weak in the throws, spends those two weeks throwing. You train the same way. After each live case, score yourself 1 to 5 on the five events: the Opening Cutscene, the Quest Map, the Number Cruncher, the Idea Hydra, and the Chairlift. Take the two lowest scores and drill those events for the next seven days.

■ OUT LOUD ONLY

Silent drills build MP and earn no XP. Say the equation before you touch a number, say each step, say the loot. Your partner hears the ramble past three minutes that you cannot.

■ THE THREE TIERS

Foundations, about two weeks. In a Foundations week you watch two full cases and time each phase, take one case and accept a rough result, rebuild its Quest Map the next day,

memorize the Fraction Scroll and the US and Utah Atlas blocks, and read one framework tree and one Codex page a day. You clear the tier once you have five full live cases in the log.

Intensify, about four weeks. In an Intensify week you take two or three cases and give one, run a blended prompt (market entry with a pricing twist), drill the Spellbook against a timer, run three Sizing Dungeon prompts and two Idea Hydra prompts, and tell one Character Select story to a partner who times it. You clear the tier with ten out-loud cases logged and a fit story you can tell in two minutes.

Adaptability, the last two weeks. In an Adaptability week you take cases from partners new to you, ask for different firm styles (a snappy Bain pace, fit-first McKinsey), state the loot from five charts in one sentence each, do one video case with hands and paper on camera, and run a 45-minute mock in the Dungeon. You clear the tier when your math holds on a clock, a chart takes you one sentence to read, and an unfamiliar case type no longer rattles you.

■ THE EIGHT-WEEK SCHEDULE

WEEK	TIER	LIVE CASES (OUT LOUD)	DRILLS	FIT STORIES
1	Foundations	Watch 2, take 1	Spellbook 10 minutes a day, Fraction Scroll	8 story seeds
2	Foundations	Watch 2, take 1	Atlas US and Utah, 1/80 trick, profitability tree	Write 2 stories
3	Intensify	Take 2, give 1	Market entry and growth trees, 3 sizings	Write 2 more, tell 1
4	Intensify	Take 3, give 1	M&A and pricing trees, 1 blended prompt, 2 hydras	Tell 2, timed at 2 minutes
5	Intensify	Take 3, give 1	5 exhibits out loud, operations tree, Codex quiz	Tell 2, rewrite 1
6	Intensify	Take 3, give 2	3 sizings with true-ups, 3 hydras	20-minute Character Select mock
7	Adaptability	Take 3 with new partners, 1 by video	Math sprints, 5 charts, 5 hydras at 4 minutes	Refine the 2 weakest
8	Adaptability	2 full 45-minute mocks in the Dungeon	Light drills, Game Over screen review	All stories once

Taken cases sum to 18, 11 of them inside Intensify, past the ten the coaches set.

■ THE TUESDAY PRACTICE

The BYU MBA Strategy and Consulting Association runs case practice on Tuesdays at 5 pm. Block all eight Tuesdays now. Second-years who recruited last year run the room. Bring one case you can give, printed with its exhibits, and name the two events you want feedback on.

■ GIVING A CASE TO A PARTNER

Playing Player 2 earns XP too: you hear a dodge from the other side and see a map from above. Pick a case with an answer key. Read the prompt once at a normal pace and keep the paper. Answer questions about the client's business and its objective. Dodge analysis in disguise with "we do not have that, but include it in your structure." Give a number only after your partner says why they want it. Note the time at each phase change. Push back once on the recommendation. Hold the teaching until the end. Coaching mid-case trains neither of you. Then give feedback in the three buckets the firms grade: structure, problem solving, and communication. Offer one keep and one fix per bucket, each tied to a timestamp.

■ TAKING FEEDBACK

Write the feedback down before you speak. Ask for the moment it happened if the note is vague. Skip the defense and spend that time on the drill that stops the ramble. Target the note that costs the most points in your next case. Keep a log of date, partner, case type, scores, and the note. Weigh the source: a second-year with 20 to 30 cases in both chairs sees patterns, a first-year partner sees timing and numbers, and an AI partner sees a strong candidate no matter how you did. Repeat the case type until you run three cases without hearing that note.

■ SCOUT THE FIRM

Before a round, learn two things about the firm across the table: the pace it likes and the shape its cases take, interviewer-led or candidate-led. Candidate forums carry the pace, the firm's own prep materials carry the shape and its sample cases, and a second-year who interviewed there last fall carries both. You are calibrating the timing benchmarks in Level 2, the 2:30 head down and the 2:15 to 3:00 walkthrough, to the firm in front of you. Walk in knowing which end of that range your Player 2 expects.

17 TROPHY ROOM

You unlock each of these fifteen Trophies with a partner watching.



TROPHY ROOM	
 ASKED FOR THE PAUSE: REQUEST TWO MINUTES BEFORE THE QUEST MAP, THREE CASES RUNNING.	 SAVE POINT SET: RECAP A PROMPT WITH EACH NUMBER CONFIRMED, ONCE OVER VIDEO.
 BRAVE QUESTION: ASK WHAT THE PRODUCT DOES IN AN INDUSTRY NEW TO YOU.	 DODGE ABSORBED: GET A QUESTION DODGED AND MOVE ON WITHOUT CALLING IT DUMB.
 FULL MAP: FOUR REGIONS, TWELVE QUESTS, EACH WITH A "SO THAT," IN UNDER THREE MINUTES.	 NO OVERLAP: YOUR PARTNER HUNTS YOUR MAP FOR AN OVERLAP OR A GAP AND FINDS NONE.
 REASONABLE NUMBERS: DEFEND AN OWNERSHIP RATE WHEN PLAYER 2 ASKS WHY.	 FRACTION SCROLL: RECITE 1/2 THROUGH 1/100 WITHOUT A GLANCE AT THE PAGE.
 THE 1/80 TRICK: SIZE AN AGE COHORT OUT LOUD AT 1.25% PER YEAR OF AGE.	 STOCK TO FLOW: CONVERT AN INSTALLED-BASE ANSWER TO ANNUAL SALES BEFORE PLAYER 2 DOES.
 TRUE-UP: ADD FLEETS, RENTALS, AND BUSINESSES TO A SIZING BEFORE PLAYER 2 ASKS.	 SLIP RECOVERY: CATCH A SLIP ON A SENSE CHECK AND FIX IT IN ONE SENTENCE.
 ANSWER FIRST: NUMBER, THEN LOOT, THEN NEXT STEP, THREE CASES RUNNING.	 HYDRA TAME: ANSWER PLAYER 2'S SECOND "WHAT ELSE" WITH NEW HEADS AND NO REPEATS.
 CHAIRLIFT RIDE: RECAP, RECOMMENDATION, LEAST-SUPPORTED NUMBER, NEXT STEPS, IN UNDER 60 SECONDS.	

18 GLOSSARY

TERM	PLAIN-LANGUAGE DEFINITION
MECE	Regions do not overlap, and together they cover the whole problem.
Hypothesis	Your best early guess at the answer, and your reason to explore one region first.
Issue tree	The problem split into branches of questions, each answerable with one analysis.
So what	The meaning of a result for the client and the action it points to, the Loot in this guide.
80/20	A few causes drive most of the result, so you chase the biggest drivers first.
Boil the ocean	Analyzing everything when you could settle the question with three analyses.
Answer first	Recommendation first, reasons second, detail last.
Pyramid principle	One conclusion on top, grouped arguments beneath it, facts beneath those.
TAM	Total addressable market: annual revenue if the product reached each possible buyer.
CAGR	Compound annual growth rate: the steady yearly rate from a start value to an end value.
Contribution margin	Price minus variable cost per unit: what each unit contributes toward fixed cost and profit.
Fixed and variable cost	Fixed cost holds steady as volume moves (rent). Variable cost rises with each unit (parts, freight).
Gross margin	Revenue minus the cost of making the product, divided by revenue: what is left to cover the rest of the company.
Operating margin	Operating profit divided by revenue, after selling costs and overhead come out.
EBITDA	Earnings before interest, taxes, depreciation, and amortization: a rough stand-in for the operating cash a business throws off.
Breakeven	The volume where revenue covers total cost: fixed cost divided by contribution per unit.

TERM	PLAIN-LANGUAGE DEFINITION
Payback	Years for annual net benefit to repay the investment: capex divided by net benefit per year.
NPV	Net present value: future cash flows discounted to today, minus the investment. A positive result means the project creates value.
ROI	Return on investment: the gain divided by the money you put in, over a stated period.
IRR	Internal rate of return: the yearly return a project earns on the cash it ties up.
Hurdle rate	The lowest return a company accepts before it funds a project.
Terminal value	The worth of all years past the end of your forecast, rolled into one figure.
CAC	Customer acquisition cost: sales and marketing spend divided by new customers won.
LTV	Lifetime value: the profit one customer brings over the whole relationship, estimated as margin per period divided by churn.
Churn	The share of customers or revenue lost in a period.
Market share	Your sales divided by the whole market's sales, in units or dollars.
Penetration	The share of possible buyers who already own or use the product.
Willingness to pay	The most a buyer would pay before walking away.
Elasticity	Percent change in units sold per percent change in price. Above 1 in absolute terms, a price rise cuts revenue. Below 1, it adds revenue.
Synergy	Extra profit a combined company earns beyond its two parts. Count it once you can name the line item.
Workstream	One Region of the Quest Map turned into a teammate's job on day one.
Steering committee	Client executives who meet the team at milestones to review findings and decide.